EduRise
INDIA

Innovators' Science

Based on NEP 2020 with Essence of NCF 2023

5

• Kalpana Yadav

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Innovators' Science 5

Latest Edition

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Preface...

The **Innovators' Science** course for classes 1 to 8 has proposed a pedagogical framework involving a spiral learning approach. It focuses on the essential parameters, including active knowledge constructions, reflective learning and project-based STEM education.

We are aware of the varied teaching pedagogies but uncertain of the methods of implementation and assessment in our curriculum transaction at times. To support this execution, the unique feature that stands apart in each book is that each parameter is supported by a pedagogical principle with suggestions for implementation in classroom practices. One of the major focuses for NEP is to change the current education system of rote learning to evidence based and hands-on learning. NEP recognises and sets a path for a multi-disciplinary approach to education which is philosophically aligned to STEM learning.

In this series, we've always promoted learning by doing, making children active learners, helping them understand the latest concepts to be at par with the developments of the world, and developing crucial 21st century skills. In this series STEM worksheets are provided to students to enhance their STEM skills through fun experiments using the skills of ideation, research, critical thinking, iteration, observation and creativity.

Our series will help students consolidate their STEM concepts in an interactive, engaging and exploratory way.

—Publisher

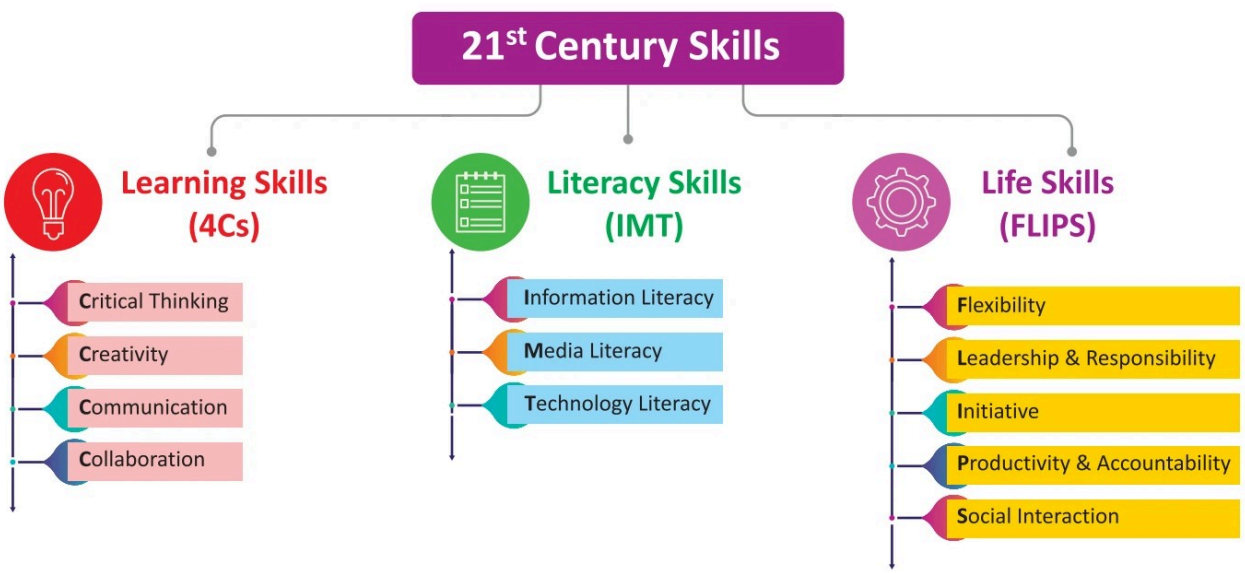
Aligned with NEP 2020 with Essence of NCF 2023

The books are crafted with new dimensions of Joyful Learning and are completely mapped with National Education Policy, 2020 with Essence of NCF 2023. In this series, new teaching pedagogies have been incorporated for better learning.

Teaching Pedagogies

- Kick Off
- Remember this
- Funtopia
- Just a Heads up
- Twisting Time
- Make Teaching Fun
- Put on your Thinking cap
- Words Aid
- Quick Review

A different skill set is required in the coming days that will enable an individual to face the challenges of real-life, leading to holistic progress. These skills are known as 21st Century Skills.



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Growing Plants

Intended Learning Outcomes

- Growing Plants
- Reproduction by Plant Parts
- Agriculture
- Seed Dispersal
- Growing Plants by Spores
- Stages of Agriculture

Kick Off

Plants are living things. They play a very important role in our lives. They not only give us food, sugar, oil, wood, fibres, rubber, gum, tea, coffee and spices but also give us oxygen for breathing.

Funtopia

Observe the pictures of some fruits given below and write 'no seed', 'one seed', 'few seeds' or 'many seeds' below each picture:

_____

_____

_____

_____

_____

_____

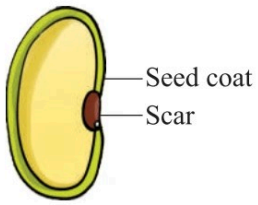
To meet our needs, we need to grow plants. Let us read how we grow plants.

Growing Plants

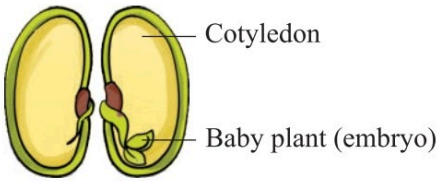
We can grow plants through seeds, stems, roots and leaves.

Reproduction by Seeds

Plants grow from young seedlings. The seedlings gradually grow into a big plant. When these plants grow old they produce new plants and ultimately die. The process by which the plants produce new plants of the same kind is called reproduction.



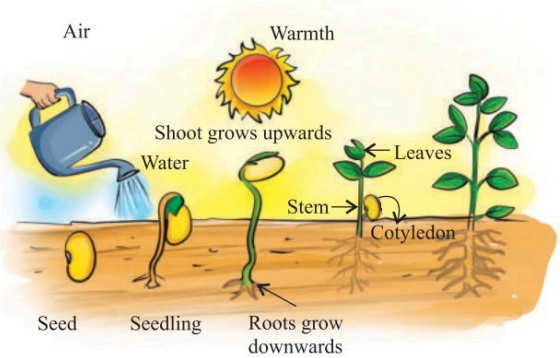
A bean seed



A bean seed splits open

Most of the plants reproduce through seeds. Seeds are present inside the fruits. Some fruits produce one seed and some produce more than one. But all of them do not grow into a new plant. What happens to them?

Some seeds are damaged by heavy rain and strong winds, while some seeds are eaten by insects and birds. Sometimes seeds do not get appropriate soil and adequate water to grow. The seeds that get proper conditions, grow into new plants. The process by which a seed grows and develops into a new plant or seedling is called germination.



Germination of seed

Just a Heads up

Some seeds like beans and peas have two cotyledons; they are called **dicot seeds**. Some seeds like maize and wheat have one cotyledon; they are called **monocot seeds**.

During germination, the seedling gets food from the seed leaves called cotyledons. The seedling uses stored food until it grows its own roots and leaves. After the food stored in the cotyledons has been used, they shrink and fall off. The seedling grows in size and develops leaves which make food for the plant through photosynthesis. The roots of the plants absorb water and minerals from the soil.

Seed Dispersal

If too many seeds are sown too close to each other, they will not be able to germinate. There will be a struggle for food, water and sunlight. Seeds need adequate sunlight, air and water to grow into healthy plants. That is why nature helps in scattering the seeds and ensures that they do not grow too close to each other. The process by which seeds are scattered away from the parent plant is called seed dispersal.

Seeds are dispersed in different ways. They may be carried far away by wind, water or animals. These are called agents of dispersal. Some plants also disperse their seeds explosively. Seed size is an important factor that determines how a seed is dispersed.

Let us understand the methods by which seeds are dispersed.

Dispersal by Wind: Some plants have lightweight seeds and hair-like structure or wings attached to them. These seeds float in the air and are dispersed by the wind. Examples, drumstick seeds, maple seeds, dandelion seeds, etc.

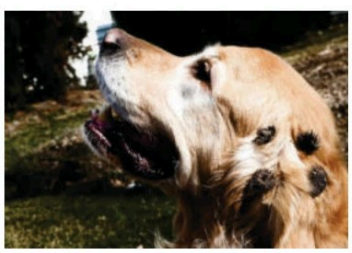
Maple seeds have wing like structures

Dandelion seeds have hair around them

Coconut seed is dispersed by water

Dispersal by Water: Some plants have spongy seeds with fibrous outer coverings, which help these seeds stay afloat in water. Water carries these seeds to large distances. Examples of such plants are lotus (spongy seed) and coconut (fibrous outer covering on the seed).

Dispersal by Animals: Animals usually eat fruits and throw away seeds. Some fruits are eaten up by birds and animals along with seeds. These seeds are then excreted in the droppings or faeces of these animals and birds. Some seeds stick to the body of animals or birds and carried away. Human beings and animals eat fruits like mango, jamun and cherries and throw away their seeds resulting in their dispersal.

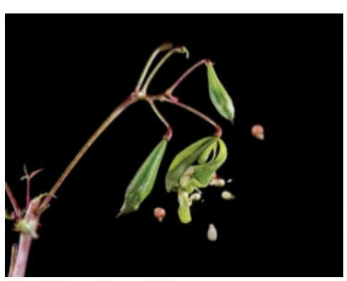


Seeds of cocklebur are dispersed by animals

Just a Heads up

In the deserts of North America, elephants eat the fruits that fall from the trees and deposit the seeds in their droppings several miles away.

Dispersal by Bursting/Explosion: Some fruits burst open when they mature. The force of bursting is enough to spread the seeds. Some examples include lady finger (okra), castor, balsam, beans, poppy, pea, etc.



Seeds of beans are dispersed by explosion

Twisting Time

Write T for true and F for false statements.

1. Plants are non-living things.
2. Plants give us nitrogen for breathing.
3. Plants grow from young flowers.
4. Most of the plants reproduce through seeds.
5. Coconut and lotus seeds are dispersed by water.

Reproduction by Plant Parts

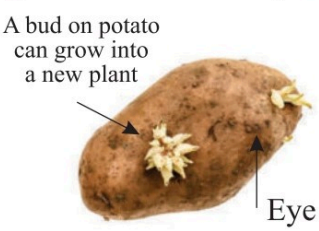
Stem, roots and leaves are known as the vegetative parts of the plants. Reproducing by stem, root or leaves of the parent plant is called vegetative reproduction or propagation. Vegetative reproduction is much quicker than reproducing through seeds.

From Stems

Some plants like rose, Hibiscus and money plant can be grown from stem cutting. The plant from which the stem is cut is called the mother plant. Stem cutting is planted in the soil. After a few days, the stem cutting grows into a new plant.

A stem is cut from the mother plant. The stem is planted in the soil;

Growing plant from stem



Potato

Potato, onion, ginger are underground stems. A potato has small buds called eyes on its surface. New plants grow from these buds. Ginger also grows from the buds on the ginger stem. Onion is a bulb-shaped stem. It will grow leaves if you plant one in the ground.

From Roots

Roots of some plants such as sweet potato, carrot, turnip, radish, and beetroot not only store food in them but also produce new plants. These roots are planted in the soil. They are known as tuberous roots.

Sweet potato can grow into a new plant

Place a sweet potato in a glass full of water and keep the glass in an open space. After a few days you will notice a new plant growing out of it.

Activity

This activity shows that some plants reproduce from roots.

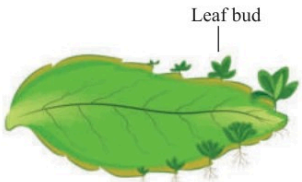
You will need: A carrot, shallow dish and water

What to do: Cut off the top portion of a carrot that has leaves. Place it in a shallow dish with the bottom part facing downwards. Pour water into the dish and place where it will receive adequate sunlight. Observe after a few days.

You will see: New leaves would have grown from the carrot top.

From Leaves

Some plants reproduce from their leaves. The leaves of Bryophyllum are thick and fleshy, and also have many buds along their edges. When the leaves fall on the soil, these buds get separated and new plants grow from these buds.



Bryophyllum leaf

Growing Plants by Spores

Spores of a fern leaf

Moss and mushrooms are examples of ferns (non flowering plants) and fungi. They do not reproduce by flowers. They produce tiny spores. Each of these spores can grow into a new plant. Fungi, protozoa, algae and mushrooms reproduce through spores.

Agriculture

Plants provide food, oxygen, medicine, timber, etc. to us. We grow plants on a large scale so that we may get enough food and other products to fulfill our needs. This practice of growing plants on a large scale for food or other products is called agriculture.

You must have seen that plants of the same kind are grown in one field. For example, in a particular field you will see only wheat plant or only rice plants. You will never find a field in which all the plants of different kinds are grown. Plants of one kind grown in a particular area at a particular time are called crops.

Crops grown in a farm

The particular area in which a crop is grown is called a farm and the person who works on a farm is known as a farmer.

In India, crops that are grown in the summer season are called *kharif*

crops. Rice, *jowar*, *bajra*, jute, hemp, peanuts, millet, maize and pulses are examples of *kharif* crops. Vegetables such as spinach, brinjal, tomato, onion, pumpkin, garlic and gourd grow well in summer season. Peach, mango, litchi, watermelon, muskmelon, plum are some fruits that grow best in summer.

Crops that are grown in the winter season are called *rabi* crops. Wheat and gram are *rabi* crops. Radish, turnips, carrots, peas, cauliflower, cabbage and beans are some vegetables that grow in winter season. Apples, oranges and pomegranates are fruits that are generally grown in the winter season.

Stages of Agriculture

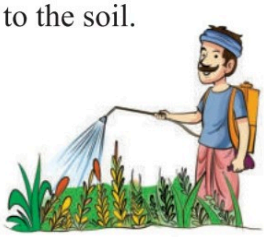
There are various stages of agriculture. These are:

The field is ploughed.

Manure is added to the soil.

After this, seeds are sown.

The field is supplied with water. This is called **irrigation**.



To protect the crops, they are sprayed with pesticides.

The crops are then cut and gathered. This is called **harvesting**.

Protection of Crops and Storage of Seeds

We need to protect our crops from:

Scarecrow

- Grazing animals like cows and goats.
- Pests such as locusts, grasshoppers and caterpillars by using insecticides or pesticides.
- Different infections and diseases in crops.
- Big animals can be kept away by proper fencing. After harvesting, grains need to be protected against moisture. They also have to be kept safe from rats, moles, birds, squirrels and insects. So, they must be stored in airtight and sealed containers.
- Plants, just like other living things, also need love and care. Under favourable conditions, they grow well. A good crop adds to the wealth of a country.

Make Teaching Fun

We have plants inside our houses too. These indoor plants become members of our family. How will you take care of them?

Words Aid

- **Germination:** the process by which a seed grows into a new plant
- **Cotyledons:** part of the seed that contains food for the baby plant
- **Dispersal:** scattering of seeds away from the parent plant
- **Agriculture:** the science of growing crops
- **Irrigate:** to provide water to the crops
- **Pesticides:** chemicals that are used to kill insects and fungi that damage plants
- **Seedling:** a new plant that has grown from a seed
- **Manure:** plants' and animals' wastes used for enriching the soil with nutrients
- **Kharif crops:** crops that grow in summer season
- **Rabi crop:** crops that grow in winter season

Quick Review

- Plants grow from young seedlings.
- The process by which the plants produce new plants of the same kind is called reproduction.
- Germination is the process by which a seed grows and develops into a new plant or seedling.
- During germination, the seedling gets food from the seed leaves called cotyledons.
- The process by which seeds are scattered away from the parent plant is called seed dispersal.
- Plants depend on wind, water, animals or explosion for seed dispersal.
- Plants can also reproduce through their body parts like stem, root and leaves. This kind of reproduction is called vegetative reproduction.
- Agriculture is the practice of growing plants on a large scale for food or other products.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. Seeds are present inside :

   a. leaves b. fruits

   c. flowers d. roots

2. The process by which a seed grows into a new plant is called :

   a. germination b. dispersal

   c. cotyledon d. agriculture

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)

3. The roots of the plants absorb water and minerals from the :
   a. leaves b. flowers
   c. root d. soil
4. Which of the following seeds is dispersed by water?
   a. Xanthium b. Coconut
   c. maple d. mango
5. *Jowar* and *bajra* are the crops of :
   a. rabi b. kharif
   c. seasonal d. none of these

3. (a) (b) (c) (d)
4. (a) (b) (c) (d)
5. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

eyes, spores, fleshy, minerals, farm

1. The leaves of Bryophyllum are thick and_____.
2. _____are found on the underside of a fern leaf.
3. The person who works on a _____is known as a farmer.
4. A potato has small buds called_____.
5. The roots of the plants absorb water and _____from the soil.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Plants play a very important role in our lives. ☐
2. The process by which the plants produce new plants is called seedling. ☐
3. Some flowers are damaged by heavy rain and strong wind. ☐
4. The leaves grow in size and develop fruits. ☐
5. Vegetative reproduction is much quicker than reproducing through seeds. ☐

D. Name the following :

1. Two plants that are grown from roots. _____ _____
2. Two plants that are grown from stems. _____ _____
3. Two seeds that are dispersed by explosion. _____ _____
4. Two examples of kharif vegetables. _____ _____
5. Two examples of rabi fruits. _____ _____

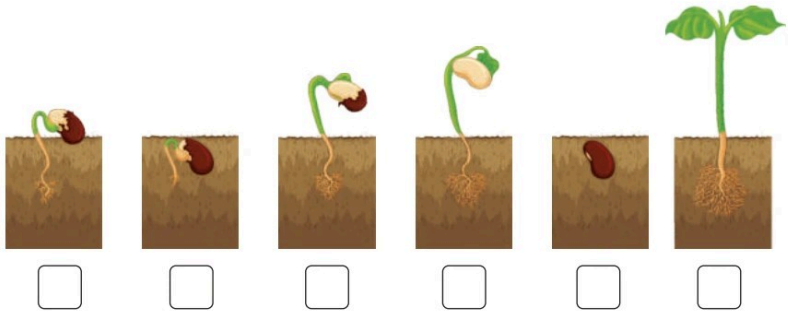
E. Answer the following questions :

1. What is germination?
2. What is reproduction?
3. What is the difference between rabi and kharif crops?
4. What is the importance of seed dispersal?

5. How do animals help in seed dispersal?
6. How are cotton and maple seeds dispersed?

Sci-Fun

Number the different stages of germination in the correct order from 1 to 6.

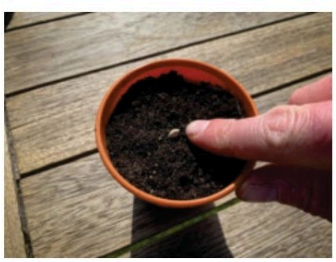


Brain Teaser (HOTS)

- We cannot grow wheat crops in summer (June or July). Why?
- What will happen if you overwater a plant or forget to water it regularly?

Subject Enrichment

Take a container and fill it with soil. Take some sunflower seeds and plant the seeds inside the soil. Place the container somewhere in warm place. Sunlight is good but try to avoid too much direct sunlight, a window sill is a good spot. Keep the soil moist by watering it every day (be careful not to use too much water). Observe them after a week. You will see that tiny roots will start to come out of the sunflower seeds. Take care and after a few days you will notice that it starts to grow into a plant.



Enhance your Life Skills

- **For the growth of plants, farmers add fertilizers to the plants. Try to know about the constituents of fertilizers and how they are helpful in the growth of plants.**
- **Why is planting trees important for a city?**

Subject Connect

Geography

Agriculture has changed the way of living. Collect information about different types of crops grown in different parts of India such as in Maharashtra, Kerala, Himachal Pradesh and Assam. On a map of India mark these states and the crops that grow best in the particular state.

2 Food and Health

Intended Learning Outcomes

- Components of Food
- Healthy Habits
- Vaccination
- Balanced Diet
- Diseases

Kick Off

Nutrients are the supply of food required by our body. Deficiency of the nutrients in our body can cause some diseases. By having some prevention we can save ourselves from the diseases. Let's know about these nutrients, and diseases caused by the deficiency of the nutrients.

Funtopia

Look at the pictures and name them.

_____ _____ _____ _____

- Which of these has a bitter taste? _____
- Which of these has a sour taste? _____
- Which of these can be eaten raw? _____
- Do you think these different food items contain different nutrients? _____

As you know we get food from both plants and animals.

Components of Food

When we feel hungry, we get food. Food gives us energy to do all our work. It also helps us repair damaged cells, grow, and reproduce.

The substances present in the food which help us to remain healthy are known as nutrients.

There are five main nutrients in the food: carbohydrates, fats, proteins, vitamins and minerals. Our body needs all these nutrients in the right amounts.

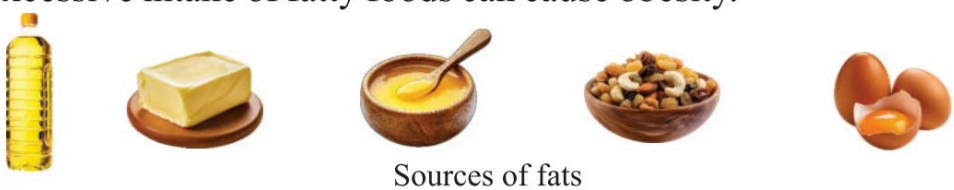
Carbohydrates

Carbohydrates provide us a lot of energy to work and play. Energy-giving food is rich in carbohydrates. Sugar, potatoes, cereals like rice, wheat, etc. are some examples of rich sources of carbohydrates.

Sources of carbohydrates

Fats

Fat is important for the human body. It also provides us energy and it helps the body use vitamin A, D, E and K. These vitamins are fat-soluble, meaning they can only be digested, absorbed and transported with the help of fats. Fat is also vital for maintaining healthy skin and hair. Rich sources of fats are cooking oils, butter, ghee, nuts, egg yolk, etc. Excessive intake of fatty foods can cause obesity.



Sources of fats

Proteins

Our body requires proteins in large quantities. They play a vital role in our growth. They help in building bones, muscles and other parts of the body. They also help in repairing injured tissues when we get hurt. Milk, eggs, meat, curd, soya bean, fish and pulses are rich in proteins.

Sources of proteins

Just a Heads up

A person who does not eat any animal product is called a vegan. Vegan diet includes all grains, fruits, vegetables, legumes and beans.

Vitamins and Minerals

Vitamins and minerals help in the normal functioning of our body. They help in protecting our body against diseases.

Fruits, vegetables and milk are some sources of vitamins and minerals. Vitamin C helps to keep the gums healthy. Calcium makes our bones and teeth strong.

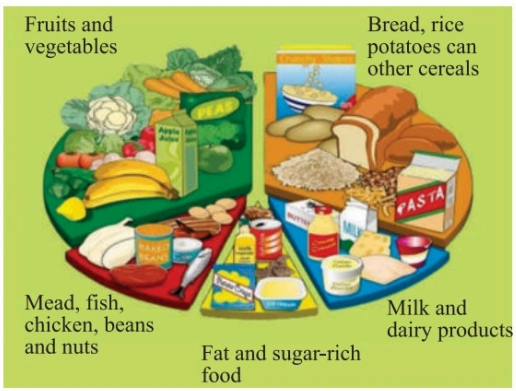
Water and Roughage

Roughage is present in fruits, vegetables and in the outer cover of the grains. Roughage is fibrous food. They are an important part of our food. They help us to get rid of the undigested food easily and also retain water in our body. *Dalia* (broken wheat) and salad are excellent sources of roughage. Water also helps to get rid of wastes from our body in the form of urine.

Balanced Diet

The food that we eat constitutes our diet. A diet that has all the nutrients in the right quantity is called a balanced diet. It is important to have a healthy diet to stay healthy and energetic.

The best way to understand the suitable amount of food items to eat on a daily basis is by looking at the food group every day.



Components of a balanced diet

In the figure of a balanced diet, each partition represents a different food group.

All of us do not need the same amount of nutrients. The amount of nutrients required by a person depends on their age, gender and the nature of the work they do.

Healthy Habits

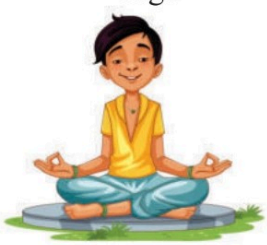
Balanced diet, exercise, correct posture and rest are essential for keeping ourselves healthy. Exercise improves our blood circulation and tones up our muscles. Running, jogging, cycling, aerobics and yoga are some forms of exercises. `

Posture is the position in which we hold our body when we sit, stand or move. We must keep our back straight while sitting or standing. Proper rest and sleep relax our muscles and give our body much-needed rest. Children need 8-10 hours of sleep in a day.

Running

We should exercise daily.

Yoga



We should sit straight.

We need proper sleep.

Twisting Time

Give one word for the following:

1. Which mineral is present in our bones? _____
2. Which type of components do fruits have? _____
3. What helps in repairing injured tissues? _____
4. What is vital for maintaining healthy skin and hair? _____

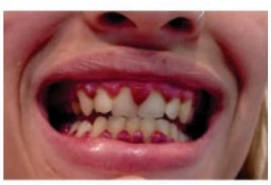
Diseases

Disease is an abnormal condition that weakens the body, and a person cannot perform regular work. Diseases may occur due to the following causes:

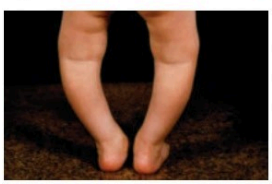
1. Deficiency of a particular nutrient in body.
2. Malfunctioning of a particular part of body.
3. Infections caused by microorganisms.

Diseases Caused by Lack of Nutrients

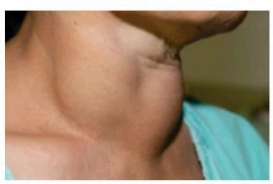
Some of the diseases are caused by deficiency of a particular nutrient in our body. Such diseases are called deficiency diseases. For example, night blindness, scurvy, rickets, goitre, etc. The disease which cannot be transmitted from person to person is called non-communicable disease.



Scurvy



Rickets



Goitre

Deficiency diseases

Some deficiency diseases and their symptoms are given in the table below:

Table : Deficiency diseases

Deficiency diseases	Caused by deficiency of	Symptoms	Foods rich in the vitamin/Mineral
Night blindness	Vitamin A	A person is not able to see in dim light	Green leafy vegetables, carrot, papaya. mango, milk and egg yolk.
Beriberi	Vitamin B1	Person has weak muscles and nerves and gets cramps	Green leafy vegetables, carrot, papaya, mango, milk and egg yolk.
Scurvy	Vitamin C	Bleeding of gums and swelling of joints	Apple, orange, lemon, tomato and *amla* (citrus fruits)
Rickets	Vitamin D	Soft bones and bow legs	Milk and fish. Sunlight helps our body to make vitamin D.

Goitre	Iodine	Swelling of a gland in neck region	Iodised salt and seafoods.
Anaemia	Iron	Person gets tired easily and looks pale	Spinach, brinjal, jaggery, apple, guava and meat.

Communicable Diseases

The diseases which can spread from one person to another are called communicable diseases. They spread through living or non-living agents.

Communicable diseases can be caused by pathogens or microorganisms like bacteria, virus, protozoa and fungi.

Some deficiency diseases and their symptoms are given in the table below:

Table : Communicable diseases

Germs	Bacteria	Virus	Protozoa	Fungi
Diseases caused	Plague, tuberculosis, typhoid, pneumonia and cholera	Common cold, polio, measles, chickenpox, flu and AIDS	Malaria, *Kalaazar* and amoebic dysentery	Infections of skin and athlete's foot

How Communicable Diseases Spread

Communicable diseases spread from one person to another through direct contact, air, water and food.

Through direct contact: The clothing and other articles used by a sick person carry germs. When a healthy person comes in contact with these articles, the germs may enter their body. These germs then may cause diseases. Chicken pox, measles, ringworm, whooping cough and common cold are spread by direct contact.

Through air : When a sick person coughs, sneezes or spits, germs are released into the air. When a healthy person takes in this air, germs can enter their body and make them sick. Therefore, we should always cover our nose and mouth with a handkerchief while sneezing or coughing. Diseases like common cold, flu and tuberculosis spread through air.

Cover your mouth with a handkerchief while sneezing

Just a Heads up

Male mosquitoes suck nectar from flowers whereas female mosquitoes suck blood from humans and animals.

Through Insects : Some insects suck blood. When a mosquito bites a person suffering from malaria, it sucks blood that contains malaria-causing germs. When this mosquito bites a healthy person, the germs are passed on and the infection spreads. Malaria is caused by a protozoan that is carried by the female Anopheles mosquito. Dengue is spread by the bite of Aedes mosquito. Plague is spread by the bite of fleas carried by rats.

Through infected food and water : Sometimes, food and water may get infected with germs, if they are not stored or handled properly. Flies and cockroaches can contaminate our food with germs. Diseases like cholera, typhoid jaundice and polio spread through infected food and water.

Female anopheles mosquito

Prevention of Communicable Diseases

We can reduce the risk of infectious diseases if we understand how they spread and how we can protect ourselves. You must remember the following things.

- Always wash your hands before or after preparing or eating food, and before and after using the toilet.
- While coughing and sneezing, cover your nose and mouth with a handkerchief or tissue paper.
- Vaccination or immunization reduces the chances of catching infection.
- Stay at home when you are suffering from an infectious disease.
- Do not share personal items like towel, comb and clothes with others.
- Wash fruits and vegetables well before cooking or eating them.
- Food must be kept covered to prevent flies sitting on it.
- Drink clean water. It can be made clean and germ free by boiling, filtering and chlorinating.

Vaccination

Vaccination is another way of preventing communicable diseases. Vaccines are available to prevent diseases like polio, cholera, diphtheria, tetanus and hepatitis. Vaccination helps our body to acquire immunity against certain diseases.

A child taking polio vaccine

Table : Symptoms and preventive measures of communicable diseases.

Communicable diseases	Symptoms	Preventive measures
Malaria	Fever, constant shivering	Avoid stagnant water puddles around your house.
Tuberculosis	Cough, fever chill, weakness, weight loss	Vaccination (BCG Vaccine).
Conjunctivitis (pink eye)	Redness of the eye, tearing, and discharge of pus	Never touch your eyes with dirty hand, never share towels and handkerchiefs.
Typhoid	Headache, fever, upset stomach	Maintain proper sanitations. Ensure safe drinking water and sanitary disposal of excreta.

Make Teaching Fun

Ask your parents to help you prepare a chart of all vaccines that you have taken from the time of birth till now.

Words Aid

- **Nutrients:** materials in food that help us to remain healthy
- **Balanced diet:** food containing various important nutrients in required amounts
- **Disease:** any malfunctioning or the deviation of body from normal functioning
- **Deficiency diseases:** diseases cause by less or no intake of important nutrients
- **Proteins:** nutrients that help in the growth and repair of the body
- **Roughage:** the fibrous part of plant foods that helps to remove wastes from the body

Quick Review

- Food gives us energy to do all our work.
- Nutrients present in food are carbohydrates, fats, proteins, vitamins and minerals.
- Water and roughage are essential for the body.
- A diet that has all the nutrients in the right quantity is called a balanced diet.
- Disease is a condition that affects the health of a person.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. There are _____ main nutrients.
   a. two b. three
   c. four d. five
2. Which of the following gives us instant energy?
   a. Proteins b. Fats
   c. Carbohydrates d. Vitamins

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)

3. Oil and butter are rich in _____.
   a. minerals b. fats
   c. carbohydrates d. proteins
4. Which vitamin is essential to prevent night blindness?
   a. Vitamin A b. Vitamin B
   c. Vitamin C d. Vitamin D
5. _____ is caused by deficiency of vitamin C.
   a. Rickets b. Beriberi
   c. Scurvy d. Goiter

3. (a) (b) (c) (d)
4. (a) (b) (c) (d)
5. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

obesity, calcium, food, nutrients, proteins

1. _____gives us energy to do all our work
2. Our body needs all the _____in right amounts.
3. Excessive intake of fatty foods can cause_____.
4. _____ play a vital role in our growth.
5. _____makes our bones and teeth strong.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Energy giving food is rich in proteins.
2. Fats help in building bones and muscles.
3. Roughage helps us to get rid of the undigested food easily.
4. Children need 8-10 hours of sleep in a day.
5. Vitamins are the abnormal condition in the body.

D. Name the following :

1. The disease caused by deficiency of Vitamin B1. _____
2. It improves our blood circulation and tones up our muscles. _____
3. The undigested part of plant food. _____
4. It is a disease caused by virus. _____
5. The insect which spreads malaria. _____

E. Answer the following questions :

1. How are carbohydrates useful to our body? Write two sources of carbohydrates.
2. What is the use of vitamins and minerals in our body?
3. What is a balanced diet?
4. What are deficiency diseases? Name three deficiency diseases.
5. Describe any two ways in which communicable diseases spread.

Sci-Fun

We obtain food from both plants and animals. Picture of some food items are given below. Write the sources from which these food items are obtained.

_____ _____ _____ _____ _____ _____

Brain Teaser (HOTS)

A child suffered from measles, and his parents did not send him to school for seven days. What could be the reason?

Subject Enrichment

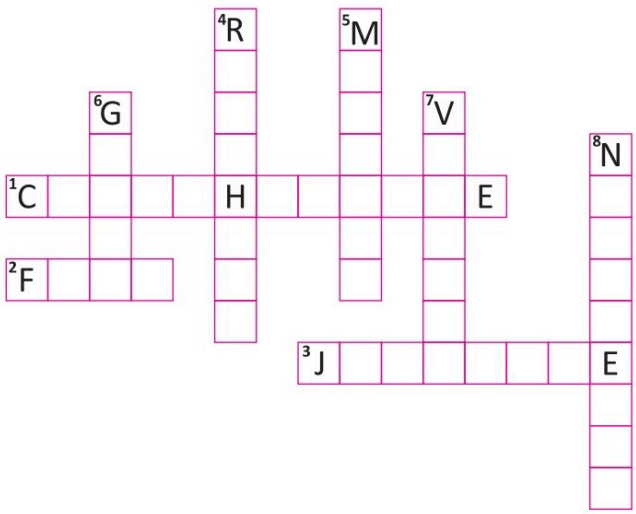
Solve the crossword puzzle by using the following clues:

Across :

1. An energy-giving food
2. A protein-rich food
3. A water-borne disease

Down :

4. Fibre present in food
5. A disease caused by mosquito bite
6. Very small living organisms
7. A protective food
8. Substance present in food



Enhance your Life Skills

Tick (✓) the correct actions you should follow to maintain good hygiene :

- Wash your hands before and after preparing or eating food
- While coughing and sneezing do not cover your nose and mouth
- Stay outside your home when you are suffering from infectious disease.
- Food must be kept covered to prevent flies sitting on it.

Subject Connect

English

Write a letter to your younger brother or sister explaining the need to eat all kinds of fruits and vegetables regularly. Use at least three proverbs related to food such as 'An apple a day keeps the doctor away'.

3 Life of Animals

Intended Learning Outcomes

- Insects
- Fish
- Amphibians
- Reptiles
- Birds
- Mammals
- Migration (Relocation) in Animals

Kick Off

Animals are found everywhere on the Earth. The place where a particular animal lives, eats, grows and reproduces, is called its habitat. The surroundings that provide animals with the conditions to live and grow are called their environment. Let's read about animals and their characteristics.

Funtopia

Read the following and tick (✓) the correct option.

1. Which of these animals live in water?
   Rat ☐ Fish ☐
2. Animals like cats, dogs, cows and human beings move with the help of their:
   feather ☐ limbs ☐
3. A whale lives in an :
   pond ☐ ocean ☐
4. Fish have neither legs nor :
   wings ☐ tails ☐

Animals are found everywhere—on land, in water and in air. Since, animals live in different surroundings each one has its own life style. A large animal kingdom is grouped into six categories on the basis of certain features. These six groups are :

- Insects
- Fish
- Amphibians
- Reptiles
- Birds
- Mammals

They have different body features, behaviour and habits. They have special structures for feeding, breathing, movement, reproduction, growth and protection from weather and predators.

Insects

Insects form the largest group of animals living on land, water and air. They are numerous in number.

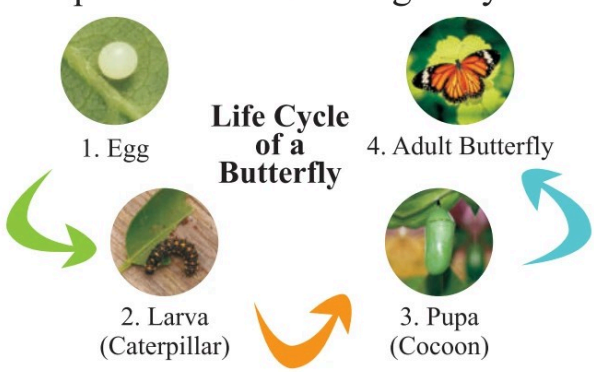
Pictures of some of the common insects are given below :

The body of an insect consists of head, thorax and abdomen. The head bears a pair of feelers, compound eyes and mouth parts suited to its method of feeding. The insects like cockroaches and grasshopper have strong biting and chewing mouth organs. Mosquitoes and bugs have piercing mouth organs for sucking blood. Houseflies have a spongy pad-like tongue. Butterflies and bees have long coiled sucking tongue or tube called proboscis.

All insects have six legs. The insects crawl by means of their legs. Ants, beetles and cockroaches use their legs for walking. A grasshopper uses its powerful hind legs to hope. The insects living in water use the legs as ears while swimming. Example : water boatman. Most insects have wings which they use to fly in the air. Insects like lice, bugs and termites don't have wings, they move by crawling.

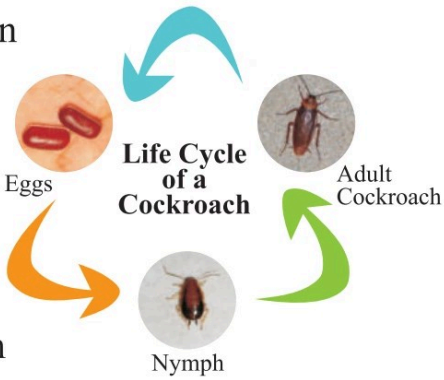
Insects like flies, butterflies and mosquitoes breathe through tiny holes in their bodies called spiracles.

Insects pass through a series of stages in their development from an egg to the adult stage. This is called life cycle or metamorphosis. The four stages in the life cycle of an insect are egg, larva, pupa and adult.



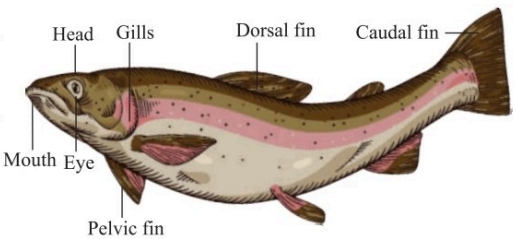
Insects, such as the butterfly, mosquito and moth undergo all these four stages in their life cycle.

Have you seen a wingless cockroach in your kitchen? It is not a wingless cockroach but the young one of an adult cockroach. It is called a nymph. Insects like cockroach, grasshoppers and termites do not undergo all the stages during their development from egg. The egg hatches into a nymph. The nymph grows directly into adult.



Fish

A fish has a boat shaped body. It has head, body, tail and wide mouth with a movable lower jaw. It swims with the help of fins. It also has a pair of eyes in the head. It breathes with the help of gills. Fish take water through its mouth. As the water passes through the gills, the oxygen dissolve in water is absorbed, and carbon dioxide is given out. Gills are red in colour due to the abundant supply of blood.



Fish are covered with a layer of flexible, overlapping scales. It reproduces by laying eggs.

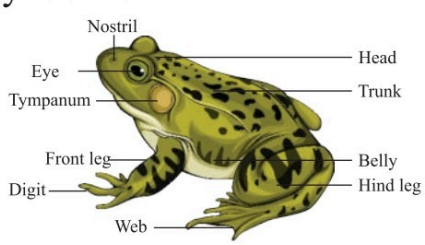
Amphibians

The animals that can live both on land and in water are called amphibians. Frogs, toads, salamanders, newts, water snakes, crocodiles, tortoises and crabs are amphibians. The skin of amphibians is smooth and without scales. The body of a frog is divided into a head and a trunk. It has a pair of bulging eyes on either side of its head.

A pair of outer membranes called tympanum, are present on the head. They are useful for hearing.

It has a wide mouth. The skin of frog is always moist.

Frogs have two pairs of limbs. The forelimbs are short. The hind limbs are long and help them in leaping on the ground. The toes of the hind limbs are webbed and help the frog to swim in water.



Sticky tongue of a frog

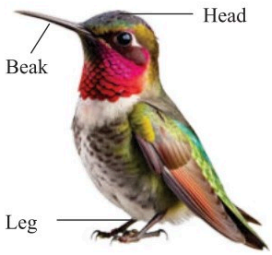
It has a long sticky tongue to catch insects. It reproduces by laying eggs. On land, it breathes with a pair of lungs. In water, the frog breathes through its moist skin.

Reptiles

Reptiles have hard, horny scales on their bodies, and they move by crawling on the ground. Garden lizards, snakes and tortoises are all reptiles. The body of a reptile is divided into the head, trunk and tail. They have a pair of limbs. Reptiles crawl on the ground. The belly touches the ground when it moves. Reptiles have long tongue to catch insects. Reptiles like snakes do not have limbs. They have scales or plates on the underside of the body that are attached to the ribs. The strong and flexible muscles of the snake contract and expand and, thus, help it to move forward.

Reptiles breathe through lungs.

Birds



Birds are the most important species among the animals. They vary in colour, size and behaviour but all of them have similar body forms. The body of the bird is divided into head, neck and trunk.

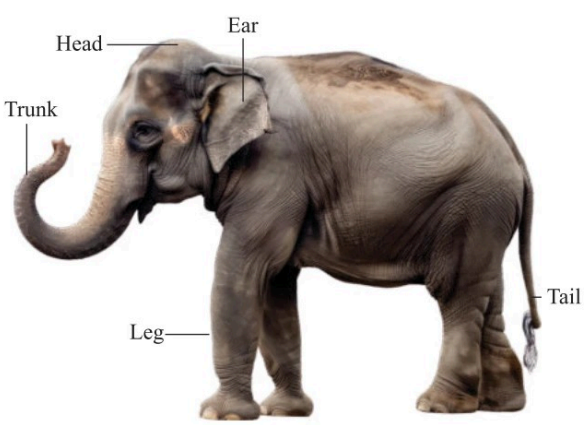
Body of birds is covered with feathers. Feathers protect them from cold, rain and help them fly. Birds breathe through lungs. The jaws of birds are modified into beaks. On land, a bird can move with the help of a pair of legs.

Birds have wings to fly. Wings are actually their forelimbs. By moving their wings up and down they push themselves through the air. Their bodies are small and light, making it easy for them to fly. Birds use their hind limbs to perch on a branch of a tree, and to walk or hop on the ground. However, not all birds can fly. Ostrich, emu and penguin cannot fly because they have large heavy bodies and very poorly developed wings. A bird lay eggs. An egg hatches into a baby bird.

Mammals

Animals that give birth to young ones and suckle them are called mammals. Mammals have hair on their bodies to protect them and to keep them warm. The body of a mammal is divided into head, neck, trunk and limbs.

All mammals and human beings breathe through lungs.



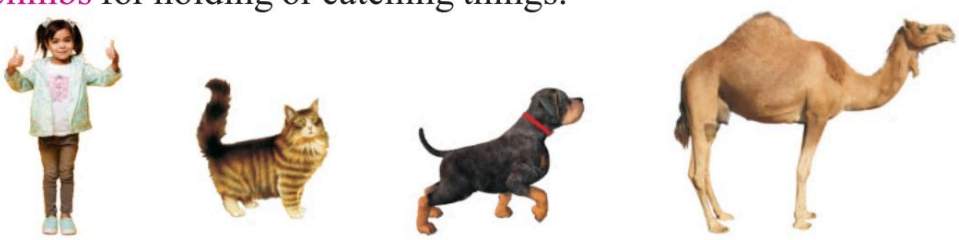
Parts of a mammal

Lungs

Humans have two lungs—one on each side of the heart. The lungs are spongy elastic-bags. The lungs have tiny tubes inside them called bronchi and a number of air pockets. The blood comes to the lungs through tiny blood vessels called capillaries. When we inhale through the nose, the air goes to the lungs which expand. Here exchange of gases takes place. The blood present in the lungs absorbs oxygen from the air and releases carbon dioxide. When we exhale carbon dioxide it is let out in the atmosphere.

Mammals have well developed organ systems and sense organs.

Mammals have four well developed limbs. Cows, camels and dogs use all their four limbs for walking and running. Human beings use only two limbs for walking, they are called hind limbs. They use their hands or forelimbs for holding or catching things.


Child Cat Dog Camel

Mammals move with the help of their limbs

Migration (Relocation) in Animals

Some animals move to long distances at certain times of the year. They travel in large groups. This mass movement of animals from one place to another is known as migration. Animals migrate to escape harsh weather, in search of food and to reach their breeding grounds where they can safely give birth to their young ones. Some birds like stork, the Siberian crane and mallard duck visit India every winter. Migration is common in birds. The Arctic tern travels a distance of nearly 70,000 km in a year from North Pole to South Pole. Scientists use various methods to study migration. Rings and tiny transmitters are fixed on animals to track their movement patterns.

Make Teaching Fun

Look for a habitat in your neighbourhood and identify different animals. Note the following.

- What are the most common animals in that habitat?
- Do they come in contact with humans?
- Where do they find their food in that habitat?

Words Aid

- **Forelimbs:** front legs in animals
- **Hind limbs:** back legs in animals
- **Habitat:** the natural home or place where an organism (animal or plant) lives, grows and reproduces
- **Migration:** the large-scale movement of animals for favourable conditions
- **Suckle:** to feed a baby with milk from the mother's breast

Quick Review

- Different animals are found in different habitats.
- The surroundings in which an animal lives is called its habitat. Forest, desert, river are some examples of habitats.
- The organs for movement, breathing and feeding in animals are suited to their habitat.
- Different animals have different breathing organs, like gills, lungs, skin and spiracles.
- Animals move in search of food and shelter with the help of fins, legs and wings.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. Insects breathe with the help of :
   a. lungs b. gills
   c. spiracles d. nose
2. Scales are found in :
   a. fish b. camel
   c. elephant c. giraffe
3. Deer is a/an _____.
   a. herbivore b. carnivore
   c. omnivore d. scavenger
4. _____ are lung breathers.
   a. Insects b. Fish
   c. Mammals d. Birds
5. _____ migrates covering a large distance.
   a. Fish b. Penguin
   c. Arctic tern d. Flamingo

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)
3. (a) (b) (c) (d)
4. (a) (b) (c) (d)
5. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

spiracles, frogs, lungs, numerous, migration, capillaries, eggs

1. Insects are _____ in numbers.
2. Insects like flies, butterflies and mosquitoes breathe through tiny holes in their bodies called _____.

3. Fish reproduce by laying _____.
4. _____ live both on land and in water.
5. The blood comes to the lungs through tiny blood vessels called _____.
6. All mammals and human beings breathe through _____.
7. Mass movement of animals from one place to another is known as _____.

C. Match the following :

Column A	Column B
1. Lion	a. Spiracles
2. Fish	b. Omnivorous bird
3. Snake	c. Without scales
4. Arctic tern	d. Strong and flexible muscles
5. Tortoise	e. Amphibian
6. Crow	f. Forest
7. Frog	g. Gills
8. Insects	h. Migratory bird

D. Tick the odd-one out giving reason :

1. Snake, Turtle, Crocodile, Lizard
2. Lion, Snake, Deer, Wolf
3. Cow, Rabbit, Rat, Squirrel
4. Kiwi, Crow, Ostrich, Emu

E. Give two examples of each :

1. Insects _____ _____
2. Reptiles _____ _____
3. Amphibians _____ _____
4. Migratory birds _____ _____

F. Give a short answer :

1. What is the natural place of living of an animal called?
2. How does a frog breathe when it is in water?
3. Why do animals migrate?
4. What are the breathing organs of a fish or prawn?
5. What are amphibians?

G. Answer these questions :

1. How is the body of the fish adapted to live in water?
2. How is a frog adapted to live both in water or on land?
3. Describe movements in mammals.

4. What features of a bird's body help it to fly?
5. Describe how air reaches the inside of an insect's body.
6. Snakes don't have legs. How do they move?

Sci-Fun

Unscramble the organs of breathing of the following animals :

1. Fish L S I L G
2. Amoeba C L L E M B R A E M M E
3. Man N S U G L
4. Insects I S R A P S C E L

Brain Teaser (HOTS)

- **How do animals know about the direction in which they move?**
- **How can humans promote the well-being of animals?**

Subject Enrichment

Match the animals that move in the same way.

1.
2.
3.

a.

b.

c.

Do all animals migrate for the same reasons?

Enhance your Life Skills

Some animals use oceanic and air currents to guide them. Birds follow the coastline to find their direction while flying. Reindeers use sunrays to find their direction. Try to know, how a bat, which has a poor vision comes to know about the right direction for flying.

Subject Connect

Art and Craft

Collect feathers of different birds and paste them on your scrapbook. Write the name of the bird under each feather.

4 The Skeletal and Muscular System

Intended Learning Outcomes

- Human Skeleton
- Joints
- Muscles
- Types of Muscles

Kick Off

Our body is made up of many organs such as heart, lungs, kidneys and the brain. Each of them performs a particular function. However, many processes in our body need a number of organs to work together. A group of organs that work together to perform a function is known as an organ system.

Funtopia

Can you tell me which organ system helps us to perform all these activities?

It is _____.

The human skeletal system or skeleton is a framework of bones that gives support to our body. It performs the following functions:

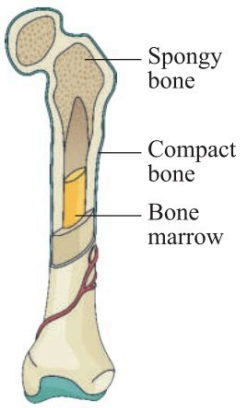
- It gives shape to our body.
- It provides support to the body so that we can stand erect.
- It protects the vital internal organs of our body such as brain, heart, lungs and spinal cord.

The major organs of the human skeletal system are the bones, the cartilage, the tendons and the ligaments. There are 206 bones in the body of an adult human being. The bones move with the help of

muscles. A new born baby has more than 300 bones. As the baby grows, some bones join together. The bones are of different shapes and sizes. All of them, however, are attached to one another.

Bone structure : Bones are made up of tissues. Bones have nerves and blood vessels. Bones are hard outside, but they are soft and spongy inside. They contain a soft tissue called bone marrow. New blood cells are produced here. Bones perform following functions :

- Bones, along with muscles, help us to move our body parts.
- Most of the blood cells are produced inside bones.
- Bones also store calcium for the body.



Structure of a bone

Human Skeleton

The human skeleton is divided into distinct parts. Let us know more:

Skull: The brain is protected by bones of the skull. There are 22 bones in the skull. Out of these, 8 bones cover and protect the brain. The remaining 14 bones make up the face. The skull is divided into two main parts—the cranium and the mandible— and that of the 22 bones in the skull, only the lower jaw bone is movable. Skull protects the eyes, inner part of the ear, nose and mouth. Skull gives a characteristic shape to our face. It changes slightly as people grow older. The male skull is slightly different from the female skull.

Human skull

Remember this

The coccyx, also called the tailbone, is present at the end of the vertebral column. It is the remnant of a tail that our ancestors must have had. A tail assists in balance and mobility in other animals. But in human beings it is almost a vestigial organ.

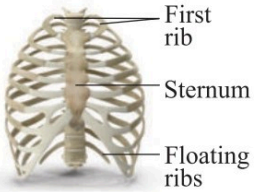
Backbone

Backbone: It is also known as the spine. The backbone is made up of 33 bones called vertebrae. They form a column called the vertebral column, which protects the spinal cord. The vertebral column forms the main axis of the skeleton. The joints between two vertebrae have an intervertebral disc. Individual vertebra cannot move. But the spine has some flexibility. That is why we can bend our back. Exercise and good posture help to maintain the flexibility of the backbone.

Remember this

Some people are born with an extra rib, which is called 'cervical rib' in the neck region. It is a type of defect that is present from birth.

Rib cage: The ribs are thin, flat, curved bones that form a protective cage around the organs in the upper body. This is known as rib cage. There are 24 bones arranged in 12 pairs. It protects the heart and the lungs. The first rib is the shortest and most curved. There is a long bone at the centre of the chest, which holds the ribs in place, known as sternum. These bones are divided into three groups :



Ribcage

1. True ribs: The first seven pairs of bones are called true ribs. 2. False ribs: The next three pairs of bones are called false ribs. 3. Floating ribs: The last two sets of rib bones are called floating ribs.

These bones are connected to the backbone at the back. The ribs protect the heart and lungs from injuries and shocks.

They protect parts of the stomach, spleen and kidneys. The ribs help you to breathe. As you inhale, the muscles in between the ribs lift the rib cage up, allowing the lungs to expand. When you exhale, the rib cage moves down again, squeezing the air out of the lungs.

Limbs: There are two pairs of limbs in our body—forelimbs (The ribcage arms) and hindlimbs (legs). These limbs help us in movement and in carrying out various activities. Our arms and legs are made up of many bones. The upper arm is made up of one bone called humerus. The lower arm has two bones—radius and ulna. The thigh or upper part of the leg has a long bone called femur. It is the longest bone in the body. It bears the whole weight of the body. The part of the leg below the knee is made up of two bones—tibia and fibula.

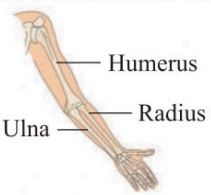
Do you think the fingers are made of one single bone or many small bones?

Remember this

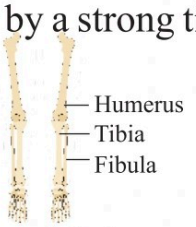
Nina Tandon
She is a biomedical engineer. She has studied biomedical engineering from Columbia University. She is a co-founder of EpiBone which is the first company in the world that grows bones for reconstruction of skeleton during disease and injury.

Girdles: The girdles are a set of bones that form a semi-circular structure to which other bones are joined. There are two pairs of girdles in human body. One pectoral girdle that forms the shoulder and the other pelvic girdle that forms the hip. The humerus bone is attached to the pectoral girdle and the femur bone to the pelvic girdle. The pelvic girdle also supports the upper body.

Bones are held together by a strong tissue called ligament.



Bones of the arm



Bones of the leg

Shoulder girdle

Hip gridle

Joints

Joints connect the bones to one another. Joints hold the bones together and allow the skeleton to move. The two bones at each joint are supported by strong band of tissues called ligaments. Ligaments are the fibrous, slightly stretchy tissues that hold one bone to another in the body, forming a joint.

Joints are of two types : movable and immovable joints.

Types of Movable Joints

There are four types of movable joints in our body.

1. Ball and socket joint: These joints have one end rounded like a ball, which fits into the cup-shaped socket of another bone. Ball and socket joint is the most movable type of joint in the human body.

   We are able to swing our arms and legs in many directions because of these joints. Such movement is called rotary. It is somewhat similar to a joystick of a computer. The hip and shoulder joints are examples of this type of joint. Let us understand the movement of ball and socket joint with an activity.

Activity

Take a soft ball and a sharp pencil. Push the pencil into the ball. Then, place the ball inside a large paper cup. Ensure that the pencil can rotate easily in all directions.

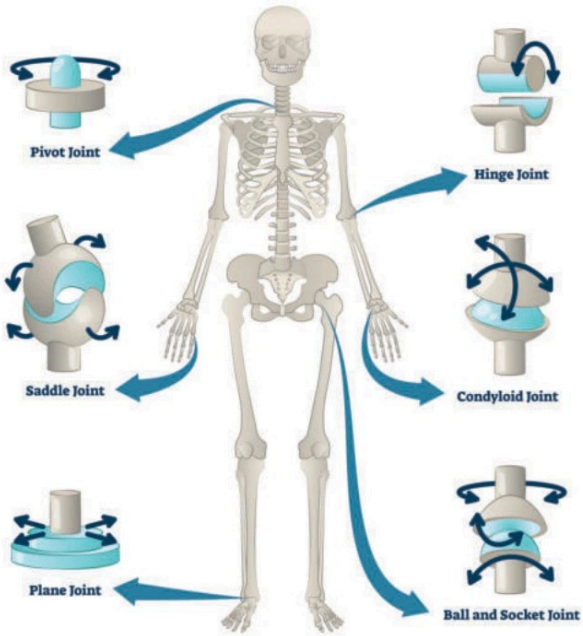
How is the movement of the ball similar to the movement of your arm?

2. Hinge joint: Have you ever noticed a door attached to a wall by metal clips called hinges? These allow the door to be opened or closed. The panel of the door opens and closes only in one direction. We have similar hinge joints in our elbows, fingers, knees and toes. Let us understand the movement of hinge joints present in our body through an activity.

Activity

Try to touch your shoulder without bending your elbow. Now, try to sit down without bending your knees. Kind of tricky, isn't it? That's because your elbows and knees are built with special joints that bend only one direction.

3. Pivot joint: In our body, the pivot joint is present between the skull and the backbone. This joint allows sideways as well as up and down movement. That is why our neck can move in all directions. It is also called the rotary joint.



Types of joints

4. Gliding joint: Our wrists and ankles have gliding joints. The bones in such a joint can glide on one another in any direction. These joints are also found in the backbone. They help the back to bend and twist. It is also called a planar joint.

Muscles

The muscles make up nearly half of the body weight in humans. The capability of movement and mobility of human body is due to the muscles. Muscles stretch throughout joints to connect one bone with another and work in groups. The muscles are attached to the bones by 'tendons'. The muscles help us to do our day-to-day activities. Muscles help to bend the arms and knees. They help to move food from one organ to another and carry out our physical movement. There are about 650 muscles in the body. Each type of muscle has a unique structure and specific role. Muscles help us to move, maintain our posture and do work with our hands. About half of our body weight is made up of muscles.

Muscular System

Just a Heads up

Both tendons and ligaments join two body parts to make the skeleton work. The difference between the two is the ligament is for three Ls—Ligament joins like to like (bone to bone). Tendon is for three Ts—tendon joins two types of parts—bone to muscles.

"Ligaments for Like to Like, Tendons for Two Types."

Types of Muscles

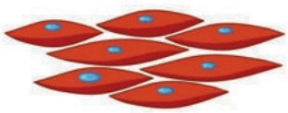
Our muscular system consists of three different types of muscles.

Striped or striated muscles: These muscles have stripes on them, just like the stripes on the body of a zebra. Therefore, they are called striped or striated muscles. These muscles are under our control. We can move them according to our will. They are also known as voluntary muscles. They are present in our arms, legs, hands and feet.

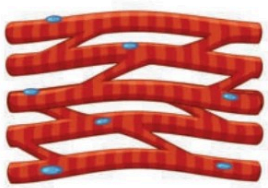
Smooth muscles: They do not have any stripes. These muscles of our body are not under our control. They are also called involuntary muscles. For example, we cannot control the muscles that are responsible for the heartbeat, movement of food or breathing. They work on their own and are controlled by the brain and the spinal cord.

Cardiac muscles: Our heart is made of cardiac muscles. These muscles are not attached to any bone. The muscles of the heart keep working 24 hours a day, throughout our lifetime. Cardiac muscles are very strong and hence, they do not get tired. These muscles are also involuntary in nature.

Striated muscles



Smooth muscles



Cardiac muscles

The bones and muscles together perform important functions of the body.

Regular exercise keeps the muscles healthy and in good condition. Correct posture also helps keep the skeletal system in proper shape.

Remember this

The Masseter muscle in the face is the strongest muscle based on its weight. It is connected to the lower jaw and cheekbone. It can snap the jaw and teeth and produce a great force equivalent to about 25 kgs on the incisors and about 90 kgs on the molars.

Working of Muscles

Some muscles in the body are involuntary, for example, muscles in the heart and digestive system. The movement of these muscles is automatic. There are other muscles in the body, which are voluntary. The brain controls working of both types of muscles. The brain sends a message through a series of nerve cells ordering muscles to contract. This is done automatically by the brain with involuntary muscles and manually by brain with voluntary muscles.

Just a Heads up

People sometimes believe that fats present in our body can be converted into muscles by exercise. This is a false perception. Fat is a different tissue from muscle. One cannot be converted into another.

Make Teaching Fun

Discuss the organ systems of the body. Show them the skeleton preserved in the biology lab. Tell them that the skeleton forms the framework of the body. Talk about the importance of bones and different joints of the body. The joints can be shown with the help of old x-ray plates. Students can be told about various exercises related to the spinal cord. Demonstrate some easy exercises in the class.

Words Aid

- **Bone marrow:** soft spongy tissue inside the bones where new blood cells are formed
- **Floating ribs:** the pair of ribs that are not attached to the breastbone
- **Involuntary muscles:** muscles that are not under our control
- **Joints:** where two bones meet
- **Ligament:** tissue that holds the bones together
- **Tendons:** tissues that attach muscles to bones
- **Vertebrae:** bones that form the backbone
- **Voluntary muscles:** muscles that are under our control

Quick Review

- Skeleton provides support to the body so that we can stand erect.
- A new born baby has more than 300 bones.
- The brain is protected by the bones of the skull.
- The backbone is made up of 33 bones called vertebrae.
- The skull is divided into two main parts.
- The pelvic girddle also supports the upper body.
- Joints connect the bones to one another.
- We are able to swing our arms and legs in many directions because of ball and socket joints.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. Which of the following helps us to keep muscles healthy?
   a. Exercise b. Sleep
   c. Rest d. Watching TV
2. Joints are places where _____.
   a. muscles meet b. bones meet
   c. tissues meet c. muscles break

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)

3. Bones are held together by _____.

a. tendons b. muscles

c. ligaments d. girdles

4. Long bones are filled with _____.

a. sponge b. simple tissue

c. bone marrow d. tendon

3. (a) (b) (c) (d)

4. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

bones, posture, skeleton, backbone, femur, blood cells, true ribs spine

1. Most of the _____ are produced inside bones.
2. _____ along with muscles, help us to move our body parts.
3. Backbone is also known as the _____.
4. Exercise and good _____ help to maintain the flexibility of the _____.
5. _____ is the longest bone in the body.
6. The first seven pairs of bones are called _____.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Bones give shape to our body.
2. Bones are soft and spongy inside.
3. There are 65 bones in the skull.
4. The spine column forms the main axis of the skeleton.
5. Sternum bones are divided into three groups.

D. Answer the following questions :

1. What functions do the human skeletal system perform?
2. What is the difference between a tendon and a ligament?
3. Why are striped muscles called voluntary muscles?
4. State the functions of bones in our body.
5. Give the differences between a gliding and a pivot joint.
6. How is a ball and socket joint different from a gliding joint?
7. Explain how muscles and bones help in the movement of our body.
8. Do you think it would be better if we had voluntary muscles in the stomach and involuntary muscles in the legs? Give reason.

Sci-Fun

Project: Make a model of the backbone

You will need some spools of thread, some large buttons and a pipe cleaner.

Thread the pipe cleaner through the spool, then a button, again a spool and then a button till you have used all the spools and buttons. The spools represent the bones or vertebrae, the buttons represent the discs between the vertebrae and the pipe cleaner represents the spinal cord inside the backbone. You have a model of the backbone!

Brain Teaser (HOTS)

What would happen if there were no bones in the body?

Subject Enrichment

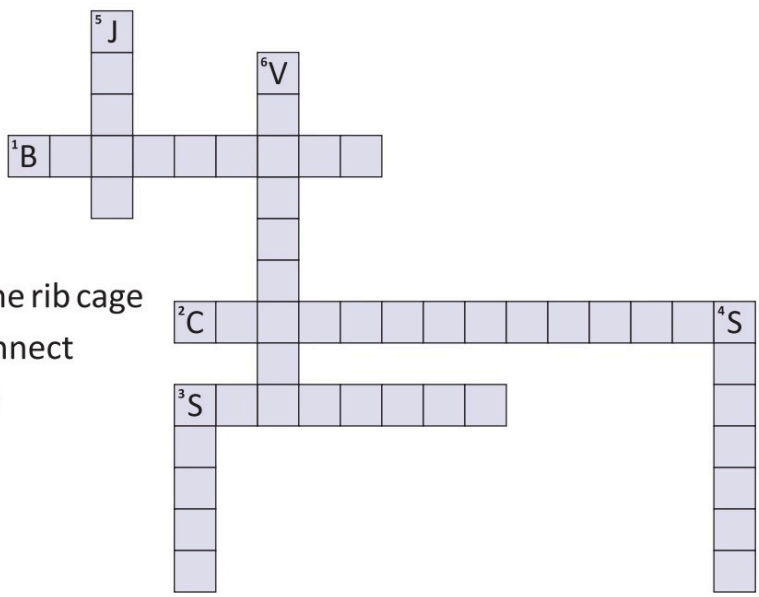
Solve the following crossword puzzle with the help of given clues.

Across

1. Soft material in the bones
2. Muscles found in heart
3. Framework of bones

Down

3. Protects the brain
4. Long bone in the middle of the rib cage
5. A place where two bones connect
6. Small bones in the backbone



Enhance your Life Skills

You must have seen people exercising in the gym. Exercise makes our bones and muscles strong.

- **Visit a gymnasium and ask the trainer which type of exercise do not make bones and muscles strong.**

Subject Connect

Sports

Muscles work in pairs. When one contracts the other expands and vice versa. Such muscles are called antagonistic muscles. Identify and name one sport where people have to work as a pair and perform work that will benefit both of them. Explain.

5 Nervous System

Intended Learning Outcomes

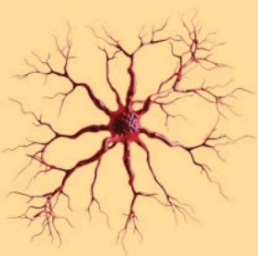
- The Nerves
- Spinal Cord
- Sense Organs
- Brain
- Reflex Action

Kick Off

What did you score in the science examination in your previous class?
What helps you to remember it? It is the brain that helps you to remember things? The brain is the most important part of the nervous system. It controls all the other body system and the sense organs.

Funtopia

Look at the pictures and write their names.



The nervous system controls all the actions and activities of our body. It consists of the brain, the spinal cord and the nerves. The brain and the spinal cord together form the Central Nervous System (CNS) which is the main centre of control in our body, and directs all our body movements. The nerves form the Peripheral Nervous System (PNS) that connects every part of the body to the spinal cord and brain. Let's study about these in detail.

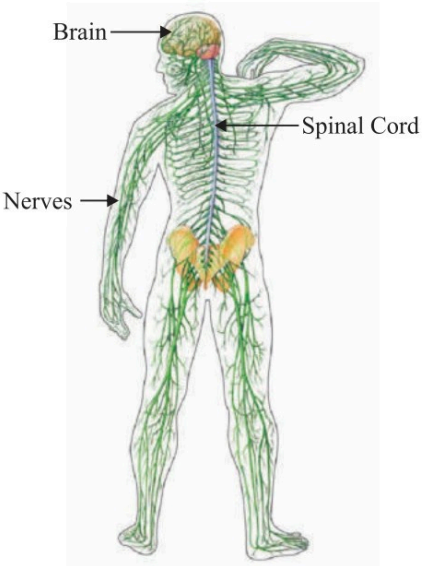
The Nerves

Nerves are made up of bundles of fibres. They spread throughout the body. They carry messages from different body parts to the brain

through the spinal cord. They also carry messages from the brain to the body parts through the spinal cord.

There are three kinds of nerves.

1. **Sensory Nerves :** The nerves which are connected to the sense organs and bring messages from the sense organs to the brain or the spinal cord are called sensory nerves.
2. **Motor Nerves :** The nerves that carry orders from the brain or the spinal cord to the muscles or glands are called motor nerves.
3. **Mixed Nerves :** The nerves that carry messages to the brain as well as bring orders from the brain are called mixed nerves.



Nervous System

Brain

The brain is the control centre of the human body. It receives information from all parts of the body, takes decisions and sends messages.

The brain is located at the top of spinal cord and is protected by the skull. The brain has three main parts :

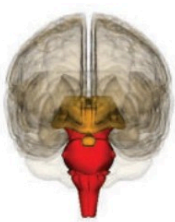
Cerebrum : It is the largest part of the brain. It has many folds so that it can fit into the skull. It controls our sense organs. It helps us to think, remember, learn, etc.

Cerebrum

Cerebellum

Cerebellum : It lies below cerebrum. It is responsible for the co-ordination of voluntary muscles. It also helps us to keep our balance and maintain the upright posture. If this part of the brain becomes defective, the balance of the body is lost.

Medulla : It is a bulb-shaped structure which lies below cerebellum. It connects the brain to the spinal cord. It controls involuntary actions like breathing and heartbeat. It works even when we sleep.


Medulla

Spinal Cord

The spinal cord is a thick cord of nerve tissue that which extends down from the brain. Messages travel along the spinal cord, to and from the brain. The spinal cord is like a connecting cable. It connects the nerves to the brain. It is well protected within the flexible but tough bony spine.

Spinal Cord

Reflex Action

Some body movements are done automatically without thinking about them. This automatic response of the body to an event is called a reflex action. The reflex action involves only the nerves and spinal cord.

For example, your mouth gets watered at the sight of ice cream and chocolate.

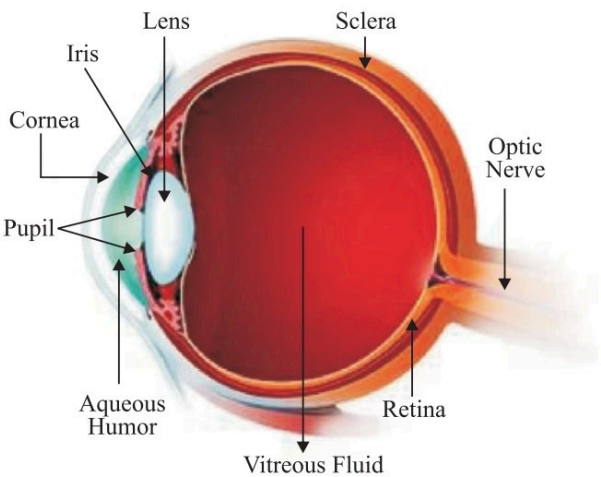
Sense Organs

The organs that connect us to the outside world are known as the sense organs. The five sense organs are eyes, ears, nose, tongue and skin.

Eyes

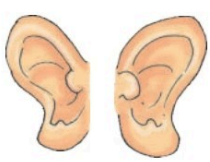
The eyes are the most important sense organs of the body. They are set in sockets in the skull. They are protected by the bones in the skull and eyelids of skin.

The coloured part of the eye is the Iris. The pupil allows light to enter the eye. The lens focuses the picture on a screen called the retina. The optic nerve carries the messages to the brain.

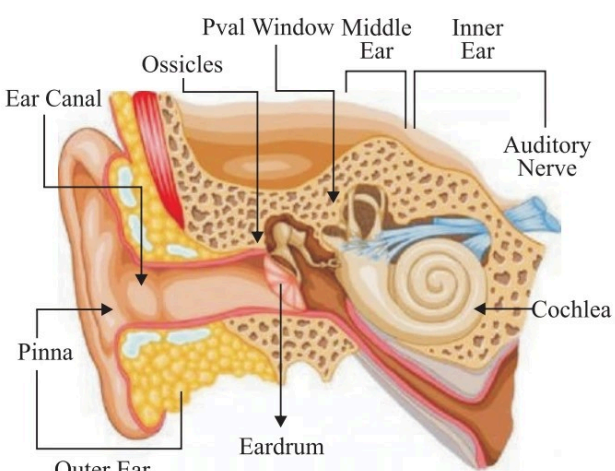


The Human Eye

Ears



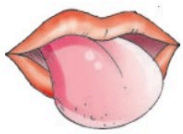
Our ears help us to hear sounds. They also enable us to keep our balance. The outer ear directs sound to the thin membrane, i.e., the eardrum, which vibrates. These sound vibrations then pass into the inner ear, where the nerves carry the messages to the brain.



Sectional View of the Ear

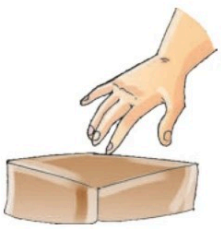
Nose

We breathe and smell through our nose. The upper part of the nose has many nerve endings which help us to smell. The sense of smell and taste are inter linked. Our food does not taste as good when we cannot smell it because of a blocked nose.



Tongue

The tongue is the main organ of taste and helps us to speak. The surface of the tongue is covered with tiny bumps. Inside these bumps are some cells called taste buds. The nerve endings in the taste buds detect the four major tastes : sweet, salty, sour and bitter.



Skin

The skin is the largest sense organ and forms a covering for the entire body surface. The five sensations that we can feel through our skin are touch, heat, cold, pressure and pain. The small pores on the surface allow the waste water of the body to come out as sweat.

Twisting Time

Match the following :

1.	Eyes	a.	Feel
2.	Tongue	b.	Smell
3.	Skin	c.	See
4.	Ear	d.	Taste
5.	Nose	e.	Hear

Words Aid

- **Cerebrum:** the largest part of the brain that controls our thinking, learning, logic, memory and intelligence
- **Cerebellum:** lower part of the brain that controls balance and muscle movement
- **Eardrum:** a thin membrane in the ear that vibrates when sound waves hit it
- **Medulla:** the part of the brain that connects it to the spinal cord
- **Motor nerves:** nerves that carry messages from the brain to the sense organs
- **Neurons:** nerve cells that form the nerves
- **Reflex actions:** actions that are controlled by the spinal cord and are very quick

Quick Review

- Nervous system carries and bring messages to and from the brain.
- The brain is the control centre of the human body.
- Medulla is a bulb-shaped structure which lies below cerebellum.
- The spinal cord is like a connecting cable.
- The coloured part of the eye is the Iris.
- The skin is the largest sense organ.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. The control centre of the human body is :
   a. nerves b. brain
   c. motor nerves d. spinal cord
2. Nerves are made of bundles of :
   a. spinal cord b. fibres
   c. tissues d. none of these
3. The coloured part of the eye is :
   a. pupil b. retina
   c. Iris d. eyelid

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)
3. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

spinal cord, cerebrum, medulla, sensory nerves

1. _____ is the largest part of the brain.
2. The _____ bring the messages to the brain.
3. _____ connects the nerves to the brain.
4. _____ controls involuntary actions like breathing and heartbeat.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. The medulla is protected inside the backbone.
2. Motor nerves send messages from the brain to the muscles and glands of the body.
3. The sense of smell and taste are interlinked.
4. The brain is located at the bottom of spinal cord.
5. The spinal cord is like a connecting cable.

D. Match the columns :

Column A	Columns B
1. Cerebrum	a. largest sense organ
2. Cerebellum	b. four
3. Motor nerves	c. carries the messages between eyes and brain
4. Optic Nerves	d. controls learning
5. Skin	e. maintains balance
6. Taste buds	f. carry messages from the brain

E. Answer the following questions :

1. Which organs make up the nervous system?
2. What are nerves? Explain different kinds of nerves.
3. What do you understand by reflex action?
4. Describe the different parts of the brain.
5. How many sense organs are there? Explain their work.

Sci-Fun

Your friend accidentally touches a hot pan and quickly pulls his hand away.

Which part of the nervous system made the hand move so fast—the brain or the spinal cord—and why?

Brain Teaser (HOTS)

Just as exercise is good for our muscular and skeletal systems, sleep is good for our nervous system. A good sleep of at least 6-8 hours is must for a healthy nervous system. However, is a part of the brain that never sleeps. Can you tell why and what would happen if it slept?

Subject Enrichment

The tongue of a chameleon is twice the length of its body. It is used to catch prey. It can hit the prey in less than a second and then draw its tongue back. Paste a picture of a chameleon in your scrapbook.

Enhance your Life Skills

What you should do to take care of your eyes. Tick (✓) the option.

1. Clean/Wash your eyes with cold water from time-to-time. ☐
2. Eat lots of fruits. ☐
3. Do not watch TV from a very close distance. ☐
4. Have sufficient light in your room when you study. ☐
5. If you face any difficulty in seeing things visit a doctor. ☐

Subject Connect

Social Studies

Study the eyes of people from different countries, for example, India, China, Russia, etc. Do you find any differences in the

- shape?
- colour of the iris?
- shape of the eyebrows?
- size of the eyeball?

Collect your findings and show it to the class.

6

Solids, Liquids and Gases

Intended Learning Outcomes

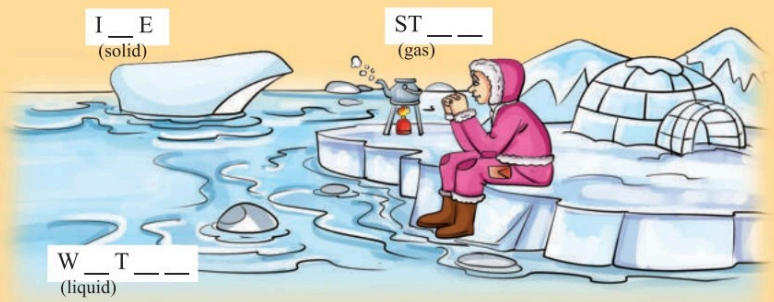
- Matter and its States
- Solutions
- Changes in Matter

Kick Off

We are surrounded by a number of things. Some of them are solids, some are liquid, and some are in gaseous form. You will see doors, windows, books, boxes, water bottles, and many other things. All these are made of different materials. So, the things around us are made up of different materials.

Funtopia

Name the three different forms of water shown in the picture.

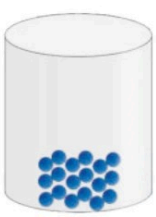


Matter and its States

Anything around us that occupies space and has mass is known as matter. Matter is made up of very tiny particles called atoms. Two or more atoms combine to form a molecule. Arrangements of such molecules in matter determine its state. Matter exists in three states: solid, liquid and gas. All the three states have different arrangement of molecules.

Solids

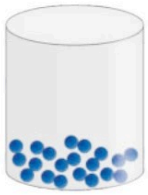
Hold your pencil tightly. Do you feel its hardness? Yes. It is solid. Things that have fixed shape, size and volume are called solids. Stones, wood and ice are examples of solids. Solids can be held. This is so because the molecules in a solid are tightly packed. They cannot move around. However, we can change the shape of some solids by using force.



Solid

Just a Heads up

There is another state of matter known as 'Plasma'. Like gases, plasma also has no fixed shape or volume and is less dense than solids or liquids. It is found in things like the Sun, lightning, and neon lights.

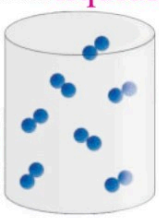


Liquids

Liquids

Milk is liquid. Is it hard? No. Can you hold it in your hand? No, it does not have a fixed shape because the molecules in a liquid are loosely packed and can move around. It can be poured into a vessel. Liquids take the shape of the vessel in which they are kept. Things that do not have fixed shape and size but have fixed volume are called liquids.

Gases



Gas

Try to catch and hold in your hand the smoke of a lighted *agarbatti*. Can you do it? No. Can you see air around you? No. Things that do not have fixed size, shape and volume are called gases.

Air is a mixture of gases. Smoke can be seen but air cannot be seen. You cannot hold gas in your hand. When air is filled into a balloon it takes up the space inside the balloon. We can pump more and more air into a football. The smell of perfume spreads quickly from one corner of the room to the other. This is because the molecules in gases are loosely packed and the gases can flow easily.

Solutions

We know that when a solute gets dissolved in a solvent, it forms a solution. When we add a small amount of sugar in a glass of water, after some time a sugar solution is formed and the sugar seems to have disappeared. But when we add sand to the water, the sand will not dissolve in water.

Substances that dissolve completely in water are called soluble substances. Salt, sugar, honey and milk are examples of some soluble substances.

Substances that do not dissolve in water are called insoluble substances. Sand, stone, buttons and seeds are examples of some insoluble substances.

Just a Heads up

Opening the door of the refrigerator for a long time lets the cold air flow out. This reduces the cooling inside the refrigerator. Thus, to save energy, we should avoid keeping the refrigerator door open for a long time and also not to open the door very often.

Most substances dissolve in water. Hence, water is known as the universal solvent. Solutions may be of different types, depending upon the state of the solute and solvent present in the solution.

Some liquids such as glycerine, alcohol and milk can dissolve in water. Such liquids that mix with each other are called miscible liquids.

Some liquids do not mix with other liquids. Add a few drops of oil in a glass of water. Stir it well. You will see that these two do not mix and the oil floats on the top of water. Such liquids are called immiscible liquids.

Changes in Matter

It is common to see ice cube changing into water when kept outside the freezer. You also know that water changes into ice when kept in the freezer. Why does this happen? The state of matter of a substance changes on heating or cooling. When ice cube is kept outside the freezer, it is subjected to heat and thus, it changes into water. The opposite phenomenon takes place when water is kept inside the freezer.

Thus, the changes are of two types: physical change and chemical change.

Just a Heads up

Alcohol also called ethanol has a freezing point of −114°C which is much lower than domestic freezers.

Physical Changes

Physical change is a change in which only the physical state of the substance changes and no new products are formed. A physical change is a reversible and temporary change. For example, on cooling, the molten wax becomes solid wax and on heating, ice can change back to water.

Melting an ice cube

Boiling water

Breaking glass

Chopping wood

Folding paper

Breaking egg

Physical Changes

A few more examples are, if a piece of paper is cut into smaller pieces it still remains paper. This is a physical change affecting only its shape and size.

Chemical Change

A chemical change is a change when a substance changes into a new substance that cannot be reversed to its original form. Thus, a chemical change is a permanent change. Burning of wax candle or burning of paper is an example of a chemical change.

Rusting of Iron

Burning Wood

Cooking an Egg

Baking a Cake

Chemical Changes

Make Teaching Fun

Take a chilled glass of water. Leave it for a few minutes. You will find water droplets around its outer surface. After some time you will notice more water collected around the glass. In this way, raindrops form and fall to the earth. Explore more about this.

Words Aid

- **Molecules:** small particles that make a matter
- **Solute:** a substance that dissolves in another substance
- **Solvent:** a substance in which the solute dissolves
- **Solution:** a mixture obtained by adding a solute to a solvent
- **Miscible:** two liquids that mix with each other
- **Immiscible:** two liquids that do not mix with each other
- **Freezing:** changing of a liquid into a solid
- **Chemical change:** a change in which a new substance is formed
- **Physical change:** a change in which only the physical properties of a substance change

Quick Review

- Anything around us that occupies space and has mass is known as matter.
- Matter is made up of very tiny particles called atoms.
- Matter exists in three states : solid, liquids and gas.
- When a solute gets dissolved in a solvent, it forms a solution.
- Water is known as the universal solvent.
- The changes are of two types: physical change and chemical change.
- In a physical change only the physical state of the substance changes and no new products are formed.
- In a chemical change a new substance is formed that cannot be reversed to its original form.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. _____ is made up of very tiny particles called atoms.
   a. Matter b. Space
   c. Shapes d. Molecules
2. Which of the following has no definite shape or volume?
   a. Solid b. Liquid
   c. Gas d. None of these

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)

3. Which of the following has a fixed volume but no fixed shape?
   a. Steam b. Oil
   c. Ice d. Wooden block
4. Which of the following is a chemical change?
   a. Ploughing field b. Melting wax
   c. Burning paper d. Folding a paper
5. Which of the following is a physical change?
   a. Baking cake b. Ploughing field
   c. Burning paper d. Boiling an egg

3. (a) (b) (c) (d)
4. (a) (b) (c) (d)
5. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box:

liquid, soluble, solids, gases, molecules

1. All the three states have different arrangement of _____.
2. We can change the shape of some _____ by using force.
3. _____ can be poured into a vessel.
4. The molecules in _____ are loosely packed
5. Salt and honey are examples of some _____ substances.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Two or more atoms combine to form a molecule. ☐
2. Substances that dissolve completely in water are called insoluble ☐
3. Water is known as the universal solvent. ☐
4. A physical change is a reversible and temporary change. ☐
5. Wax candle is an example of physical change. ☐

D. Write one word for the following:

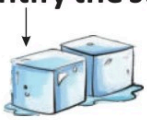
1. Small particles that make up the matter _____
2. The space in between the molecules _____
3. Two examples of physical change _____
4. A change that is permanent _____
5. Two liquids that mix with each other _____

E. Answer the following questions:

1. Define matter. How many states of matter are there?
2. Why does liquid flow?
3. What are miscible and immiscible liquids?
4. Define chemical change and give examples.
5. How can you say that baking a cake is a chemical change?

Sci-Fun

Identify the states of matter—Solid, Liquid or Gas.

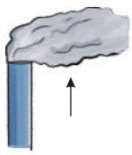
1.  _____

1. _____

1.

_____

4. _____

4.  _____

4. _____

7. _____

7. _____

7.

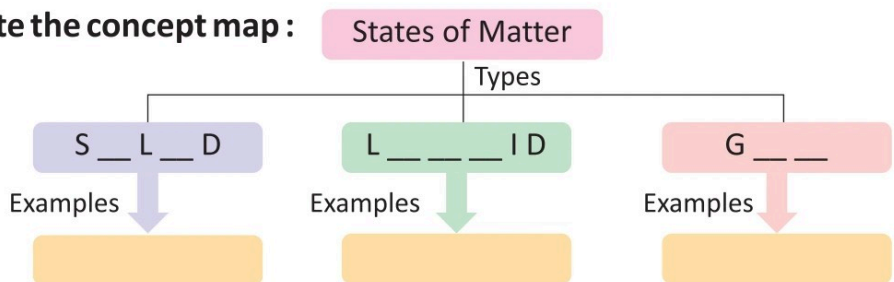
_____

Brain Teaser (HOTS)

On adding orange juice to water, the water turns orange. What could be the reason?

Subject Enrichment

Complete the concept map :



Enhance your Life Skills

- **Your mother lights the incense sticks in the *puja* room, but you get the fragrance sitting in the drawing room. How would you explain the phenomenon?**
- **Clouds are just water vapour, so why do they move?**

Subject Connect

English

Imagine a story of a drop of milk that is going to be turned into ice cream. Sugar, may be food colour and fruits will be added to it to make a lovely, delicious ice cream. Write a paragraph of about 80 words, describing its journey from being a liquid drop of milk to a spoonful of solid ice cream. Use the words given below for reference.

Solid, liquid, freezing, cooling, solution, solute, solvent

7 Force, Energy and Simple Machines

Intended Learning Outcomes

- Force
- Energy
- Types of Force
- Types of Energy
- Work
- Simple Machines

Kick Off

Force, work and energy are the terms we generally use in our daily lives. Let us perform this short exercise to find out how much we know about force, work and energy.

Funtopia

Identify the action being performed ('push' or 'pull') in each case.

a. _____

b. _____

c. _____

d. _____

e. _____

f. _____

We either push or pull objects to move them. Why do we apply force? We apply force in order to bring about some change. For example, we push forward the paddles of our cycle to enjoy the ride. A horse pulls a cart to move it. We can say that a push or pull on an object is called force.

A boy pushing the paddles of a cycle

A horse pulling tonga

Force

Power and energy make our work easy. Let us study about force, work and energy.

Force is an external factor which makes objects to change its position. A push or pull applied on an object is called force. So, force is needed to push or pull an object. A force can do the following work :

- It can move a stationary object.
- It can stop a moving object.
- It can make a moving object move faster.
- It can slow down a moving object.
- It can change the direction of a moving object.
- It can change the shape and size of an object.

Writing on paper

Opening the door

Playing football

Running for exercise

Pulling a suitcase

Playing hockey

Types of Force

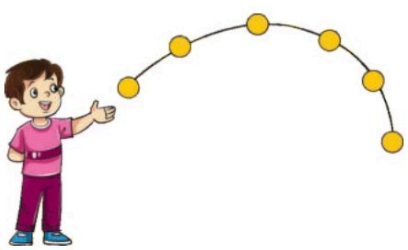
There are different types of forces:

Magnetic Force: When we bring a magnet near an iron object, say iron nails, it attracts the iron nails towards itself. This is because of the fact that magnet has a force, which attracts. It is called magnetic force.

Mechanical Force: Simple machines work on mechanical force. When we use a pair of scissors to cut cloth or use a knife to cut vegetables, it uses mechanical force for that. This force saves our energy and time also.

Muscular Force: Force applied by our muscles is called muscular force. When we push, pull or lift something, we apply muscular force.

Gravitational Force: Everything that we throw in upward direction comes down on the Earth. It is because of the force exerted by the Earth on the object. This force is called gravitational force. It is the pulling force of the Earth.



Frictional Force: When we roll a ball on the ground, it stops after some time due to the force of friction. This force slows down a moving

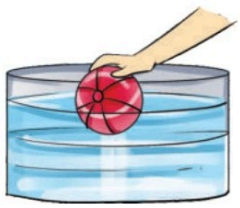
Just a Heads up

We are able to stand on the Earth's surface because of the gravitational force.

object. We are able to walk because of the frictional force. Vehicles are stopped due to the frictional force. Frictional force is also called force of resistance.

Elastic Force: When we stretch a spring, it increases in length. But when we release it, it regains its original shape or position. This is due to elastic force.

Buoyant Force: If we push down a ball floating on water, we feel an upward thrust. This upthrust or push is called buoyant force.



Work

When you read a book, you are not doing any work. But when you pull a chair, it is said that work has been done. When we apply force to move any object, we do work.

But, when you push an almirah and it does not move, no work is said to be done. Thus, work is done only when force is applied on an object and it moves. So, a work is said to be done only when the applied force causes a change in the position of the object in the direction of the applied force.

Energy

Energy is the ability to do work. We need energy to perform work. We obtain energy from nature through different means. We obtain energy from nature through different natural sources like, sunlight, wind, water, animals and plants.

We require energy to perform many activities. Playing, running, walking, all these activities need energy. Only the body which has energy in it, can produce force to carry out work.

Types of Energy

Following are the different types of energy:

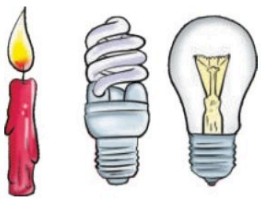
Heat Energy

When things like coal or wood are burnt, a lot of heat is produced which is in the form of heat energy. This heat energy can be used for cooking, heating, etc. Sun is the main and natural source of heat. The heat that we get from the Sun is called solar energy. It is a form of heat energy. The solar cookers and solar heaters work on solar energy.

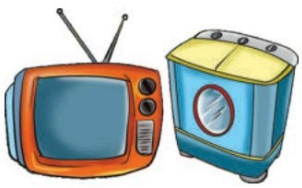
Just a Heads up

A 100 watt bulb consumes only 1 unit of energy to give light for 10 hours.

Light Energy



Energy that we obtain from light is called light energy. Bulb, candle, tube, etc. are sources of light energy. Sun is also a natural source of light energy.



Electrical Energy

Energy that is obtained from electric power or current is called electrical energy. Television, washing machine and other electrical appliances run on electrical energy.

Kinetic Energy

Energy of an object generated due to its motion is called kinetic energy. A moving car, moving truck, moving toy, etc. have kinetic energy.

Potential Energy

Potential energy is energy which is stored in an object due to its position. Water in a tank has potential energy.

Twisting Time

If a 100 watt bulb consumed 1 unit of energy to give light for 10 hours, how many units will be consumed by a geyser of 2000 watts in 10 hours?

_____

Mechanical Energy

It is the combination of both kinetic and potential energy. A ceiling fan has mechanical energy.

Wind Energy

Wind has the power to move things. This power is called wind energy. Wind energy is used to operate wind mills. Wind mills are used to generate power and electricity.

To make our work easy and fast, we use machines. There are a number of simple machines which make our work easier. Let us know more about simple machines.

Simple Machines

There are six types of simple machines:

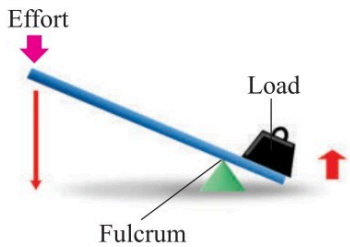
Inclined plane

Roller coaster

An inclined plane is a sloping surface, which makes it easier to move heavy objects to a certain height. Roller coaster, sloping roads, ramps, etc. are some of the examples of inclined plane.

Levers

Lever is one of the most common and simplest of all machines. They help us lift loads with lesser effort. A lever has three parts: load, effort and fulcrum. Load is the object you are lifting. Fulcrum is the point at which the lever pivots. Effort is the force applied to move the object.

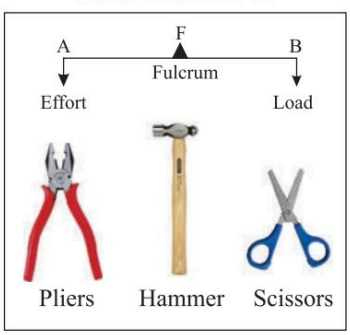


Levers are divided into three classes depending upon the relative positions of the load, the effort and the fulcrum.

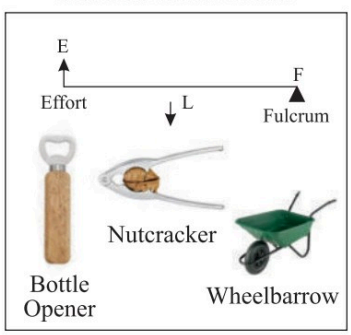
Kinds of Levers

- **First class lever:** The position of the fulcrum is in between the load and the effort. Examples of first class lever are pliers, hammer and scissors.
- **Second class lever:** The position of the load is in between the fulcrum and the effort. Examples of second class lever are nutcracker, bottle opener and wheelbarrow.
- **Third class lever:** The position of the effort is in between the fulcrum and the load. Examples of third class lever are spade, forceps, fishing rod, etc.

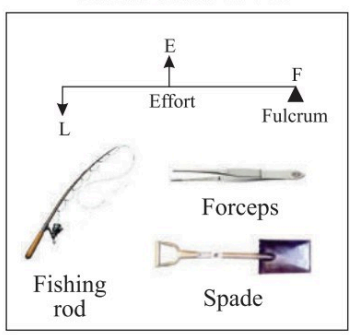
First class lever



Second class lever



Third class lever



Pulley

Sometimes, it's not easy to lift heavy objects with only human force. This problem can be solved with the help of a simple machine like a pulley. A pulley is a wheel with a fixed axle with grooves around the outside edges to guide a rope or a cable. The rope is attached to the

grooves of the wheel to pull down or lift any object. Flagpole, crane, pulley of a well to fetch water are some of the examples.

There are two types of pulleys: fixed and movable.

1. The fixed pulley is fitted to some support and is commonly seen in a pulley used for drawing water from well. The flagpole is another example of fixed pulley.

Force applied and load moves in opposite direction.

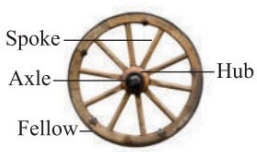
2. The movable pulley is a pulley that is free to move up and down and is attached to a ceiling or other object by two lengths of the same rope. The movable pulley reduces the effort almost to half as two parts of the rope support the weight. For example, construction cranes, elevators, etc.

Force is applied and load moves in same direction.

The only difference between the two pullies is that the rope moves through the wheel in case of fixed pulley and in movable pulley the wheel moves through the rope.

Wheel and axle

Sometimes, it is very difficult to move a heavy load due to gravitational force that pulls the object towards the ground and friction resisting the motion of the object. This problem can be easily solved by the wheel and axle. The wheel with an axle functions as an excellent machine. A wheel is locked to a central axle and they rotate each other when a force is applied to either one of them. We see the wheel and axle in many of our day to day activities. For example, steering wheel of a car, pedal of a bicycle, wheelbarrow, etc.



Screw

We use a screw many times. It is a simple machine and used to hold things together. A screw is actually an inclined plane wound around a rod. The grooves cut on it make it a more efficient simple machine. The strength of a screw's hold depends on the width of the grooves and distance between them. The screw has become an important part of the

mechanical age. It can be used to hold things together, lift heavy loads, drill holes in objects, etc. Can a screw be compared with a nail? Which works better? You will always find two pieces of wood are joined with a screw and not with a nail. The grooves on screw make the grip stronger in between the two pieces of wood.

Wedge

In a wedge two inclined planes or sloping sides are joined back to back. It is also a simple machine used to split things. Axe, chisel, knife, etc. are some of the examples of wedge. We use these things to cut and split things apart. Its edge is sharp for cutting.

Make Teaching Fun

Take different types of balls, varying in size and weight. Find out if the weight of an object affects the amount of force needed to move it by a certain distance.

Words Aid

- **Force:** a push or a pull
- **Work:** when applied force changes the position of an object
- **Energy:** ability to do work
- **Heat energy:** energy obtained by burning fuels like oil, gas, etc.
- **Sound energy:** energy produced by the vibration of objects

Quick Review

- In our day-to-day life, we do many types of work.
- Power and energy make our work, easy.
- Force is a push or pull on an object.
- Energy is the ability to do work.
- There are many types of forces and energy.
- Energy cannot be produced. It only changes from one form to another.
- An inclined plane is a sloping surface.
- A lever has three parts: load, effort and fulcrum.
- Levers are divided into three classes based on the relative positions of the load, the effort and the fulcrum.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. Fuels on burning produce _____ energy.
   a. mechanical b. light
   c. solar d. heat
2. A spring is an example of _____ force.
   a. muscular b. mechanical
   c. wedge d. elastic

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)

3. From where do we get heat energy?

a. Plastic b. Sun

c. Water d. Rain

4. Due to _____ force, a mug floats on water.

a. elastic b. frictional

c. buoyant d. axle

3. (a) (b) (c) (d)

4. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

slow, screw, force, easier

1. Push or pull is called _____.
2. Force can _____ down a moving object.
3. Simple machines make our work _____.
4. A _____ is used to hold things together.

C. Match the columns :

Column A		Columns B
1.		a. Magnetic force
2.		b. Buoyant force
3.		c. Solar energy
4.		d. Light energy
5.		e. Wind energy

D. Answer the following questions :

1. What is force?
2. Explain any four types of forces.
3. What is work?
4. When is work said to be done? Explain giving a suitable example.

5. What do you understand by buoyant force?
6. Define friction. Explain its advantages and disadvantages.
7. Differentiate between light energy and heat energy.

Sci-Fun

Your friend wants to join two pieces of wood. What will you suggest them to use a nail or a screw and why?

Brain Teaser (HOTS)

- **How did the ancient Egyptians move heavy blocks during the construction of the pyramids?**
- **Why do wheels on skates not have grooves?**

Subject Enrichment

Against the following, write 'L' if it is a lever; 'I' if it an inclined plane or 'W' if it is a wheel or axle against each of the following.

1. A pair of tongs _____
2. Bottle opener _____
3. Ramp at a car service station _____
4. A pair of scissors _____
5. Ramp at a railway station _____
6. Sewing machine _____

Enhance your Life Skills

- **Visit a circus, fair or children's special zone. Have fun taking various joy rides. Observe how each of these rides work as a different type of simple machine.**
- **How can we overcome the limitations of using renewable energy sources such as solar power?**

Subject Connect

Value Education

- **Air conditioners (AC) gives out a lot of heat and also release harmful gases. This harms the environment and causes Global Warming. Discuss the ways to reduce the use of Acs.**
- **Why is it important to conserve energy resources?**

8 Air and Water

Intended Learning Outcomes

- Air and Atmosphere
- Importance of the Atmosphere
- Properties of Air
- Ventilators in Houses
- Impurities in Water
- Separation of Soluble Impurities
- Layers of Atmosphere
- Composition of Air
- Hot and Cold Air
- Water
- Separation of Insoluble Impurities from Water
- Purifying Drinking Water

Kick Off

We need air and water to live. We mainly use air for breathing and water for drinking. Air and water have many other uses too. Both air and water are precious gifts of nature found on the Earth.

Funtopia

Look at the pictures and answer the questions that follow.

What is the girl blowing into the balloon?

What is helping the kite to fly in the sky?

Air and Atmosphere

We cannot stay alive for more than a few minutes without air. Why? We need the oxygen present in the air to breathe. Air is all around us. We cannot see it but we can feel it. The thick layer of air that surrounds the Earth is called the atmosphere. It is several hundred kilometres thick above the Earth. Without air there would be no life on Earth. As you go higher, the amount of air decreases—that is why mountaineers have to carry oxygen cylinders when they climb high up on mountains.

Layers of Atmosphere

The atmosphere has several layers. These layers are troposphere, stratosphere and mesosphere, thermosphere and exosphere.

Troposphere is the nearest layer to the Earth's surface. It extends upto 15 kms above the surface of the Earth. Most of the weather changes take place in this layer.

The stratosphere lies above the troposphere and is about 35 km thick. This is where rare jet planes usually fly.

Beyond the stratosphere lies the mesosphere. In this layer, temperature begins to fall as we go higher. That is why it is known as the cold layer.

The thermosphere is called the hot layer because it is characterised by high temperatures. Space shuttles orbit in this layer.

Exosphere is the outermost layer of the atmosphere. The air here is very thin. Beyond the exosphere is a vast, empty space. There is no air in this space.

Importance of the Atmosphere

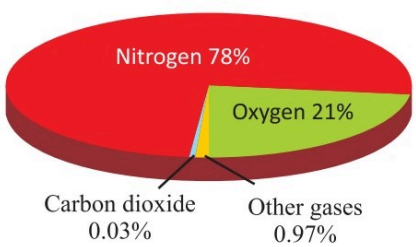
The atmosphere plays a vital role in sustaining life on the Earth.

- The atmosphere provides oxygen for breathing to all living things.
- It provides carbon dioxide to plants for photosynthesis.
- Atmosphere also traps the heat of the Sun and helps in maintaining the right temperature on the Earth.
- The ozone layer present in the atmosphere protects us from the harmful ultraviolet rays of the Sun. These rays cause skin cancer and problems in the eyes.

Composition of Air

Air is a mixture of gases. The two main contents of air are—Nitrogen and oxygen. In addition to these, air also contains small quantities of carbon dioxide, water vapour and traces of other gases. See the proportions of gases in the given picture.

Nitrogen = 78%
Oxygen = 21%
Carbon dioxide = 0.03%
Other gases = 0.97%



Composition of air

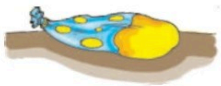
The amount of water vapour present in air is called humidity.

The humidity is high when the amount of water vapour is more in air. In humid air the perspiration from our body surface does not evaporate quickly so we feel uncomfortable. We can feel the humidity in air before and after the rain.

Properties of Air

Air has Weight and Occupies Space

Take a balloon and blow air into it. What happens to its size? You will find that blowing air into a balloon increases its size. This is because air occupies space.



Air also has weight. You can check this by comparing the weight of an inflated balloon with that of a deflated one.

Air Exerts Pressure

Because air has weight, it presses downwards. Air can push things. Moving air is called wind. Parachutes, kites, gliders, and sailing boats all depend on wind (moving air) to move.

Scooters, trucks, buses and cars are all supported on their tyres. Air is pumped into their tyres.

Air pressure too plays an important role in our day-to-day activities.

Try opening a packet of milk. When you cut a small opening on one side, milk flows out slowly. If you make another slit on the other side, milk flows out rapidly. Can you guess why? This happens because air enters from the other slit and pushes the contents out of the pack.

Air pressure helps us in many ways. If there was no air pressure you would not be able to fill up your fountain pens. Even sucking through a straw would not be possible without air. Droppers also make use of air pressure.

Syringe,

dropper, and water

drinking with the help of straw work because of air pressure.

Activity

Aim : To observe air pressure in a dropper.

Materials required : A dropper and water.

Procedure : Press the bulb of the dropper. Lower the dropper into water, keeping the bulb pressed. Now release the bulb.

What did you observe? _____

What do you think is the reason for your observation? When the bulb is pressed, air goes out of the dropper. When it is released in water, air pushes water into the dropper.

Hot and Cold Air

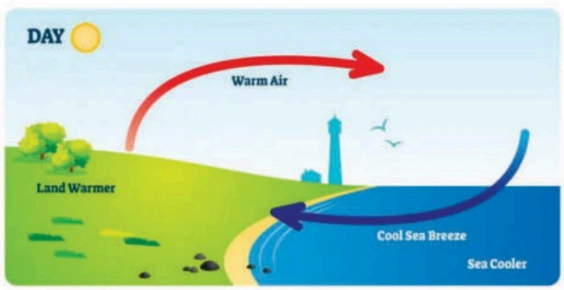
The Sun heats up land and water. The air above them also gets warm.

Heat affects land and water differently. Land absorbs and loses heat more quickly. On the other hand water absorbs and loses heat more slowly than the land. The warm air above the land and sea is always lighter and therefore it rises up. Cool air, being heavier, sinks down.

When warm air rises up, cool air rushes in to take the place of warm air. Thus air currents are formed. This process continues and causes convection currents. Land and sea breezes are caused due to convection currents.

Sea Breeze

During the day, due to the heat of the Sun, the land heats up faster than the sea. The air above the hot land also gets heated faster than the air above the sea surface. This warm air is lighter and rises up. The air above the sea is cooler. This cool air from the sea rushes in towards the land to take place of the warm air. This moving air is called the sea breeze.



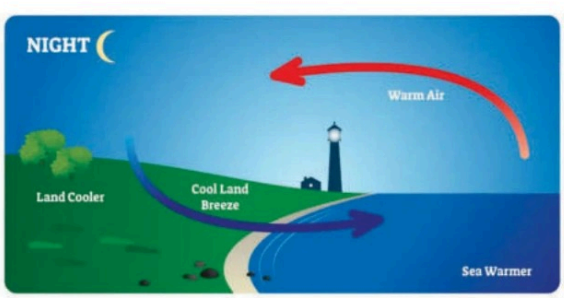
Sea Breeze

Causes of Rain and Monsoon

During summer, the land near the sea is heated more rapidly than the nearby water. The air above the land area also gets warm. The warm air rises up and causes a low pressure area. Cool sea breezes then begin to flow from the oceans towards the land to fill the space. The winds from the sea carry with them moisture. The moisture-laden winds cause rain. The monsoon winds bring heavy rainfall.

Land Breeze

At night, the land and the sea start cooling. Land loses heat faster than the sea. So the sea is warmer than the land. As a result, the air above the sea, which is also warmer, rises up. The cool air from the land moves in towards the sea to take its place. This moving air is called the land breeze.



Land Breeze

Activity

Take two bowls. Fill one bowl with sand and the other with water. Leave the bowls out in the sun for two hours. With the help of a thermometer, measure the temperature of the sand and the water. Record your reading.

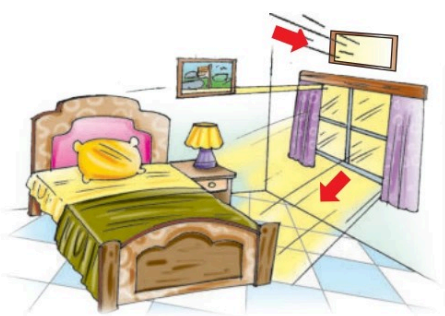
Now keep the bowls inside the room for another two hours. Again measure the temperature. Record your reading. Write your conclusion:

	Temperature of sand	Temperature of water	Conclusion
In the sun			
Inside the room			

Ventilators In Houses

We need to get enough light and fresh air in our houses. Sunlight and fresh air kill the germs and keep the house in a healthy condition. That is why, we provide ventilators in our houses. Ventilators allow fresh air to come in and let the foul air go out. Ventilators work on the principle of air currents. Warm air inside the room rises up and go out. Cool fresh air rushes in to take its place.

Even if we keep all doors and windows of our rooms closed, ventilators allow the free circulation of air in the rooms. We need fresh air to breathe in and at the same time contaminated air having gases like carbon dioxide and carbon monoxide need to be rushed out of the places where we live in. Ventilators serve this purpose.



A ventilated house

Water

Water is tasteless, odourless and colourless substance. We can feel or see water.

- A liquid that contains a dissolved material is called a solution.
- A substance that is dissolved in liquid is known as the solute.
- A substance generally liquid in which the solute dissolves is called the solvent.

Sugar (Solute)	+	Water (Solvent)	⟶	Sugar solution (Solution)
Common salt (Solute)	+	Water (Solvent)	⟶	Salt solution (Solution)

Because of its property of dissolving solids, water is used as a solvent to prepare many solutions.

Solids like sugar, common salt, copper sulphate, potassium permanganate and glucose dissolve in water. Such substances are called soluble substances.

Substances like iron nails, wood, sand, sawdust, clay, oil, wax and coconut oil do not dissolve in water. Such substances are called insoluble substances.

Insoluble substances either sink or float in the water.

If you throw a feather in a river, it remains on the surface and flows in the direction of flow of water. Similarly, if a wooden plank is thrown in the river, it remains on the surface. This phenomenon is called floatation.

Twisting Time

Name any three other things that float in water.

_____ _____ _____

Impurities in Water

The water that we use mainly comes from rivers, ponds, lakes and wells. Often sand, dirt, mud, germs, etc. make this water unfit for consumption. We call these substances impurities.

Impurities are of two kinds :

a. Soluble impurities are the ones that dissolve in water (e.g., salt).

b. Insoluble impurities are the ones that do not dissolve in water (e.g., sand and germs)

Separation of Insoluble Impurities from Water

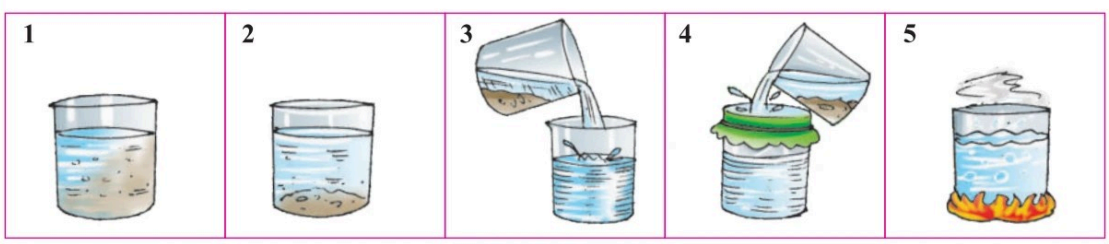
Insoluble impurities can be separated by the following methods :

Sedimentation

In this method, water is allowed to stand undisturbed for a few hours in a container. At the end of this time, the insoluble impurities settle at the bottom of the container. These are called sediments and the process is called sedimentation.

Just a Heads up

The rate of sedimentation can be improved by adding special chemicals like alum.



Separation of impurities from water

Decantation

This process consists of slowly pouring the clean water at the top into another vessel. The sediments are left behind. Decantation always follows sedimentation.

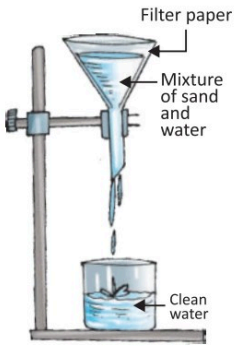
Just a Heads up

Fine cotton cloth acts like a filter. It has very tiny openings through which water can pass easily but large mud particles cannot.

Filtration

Insoluble impurities can be removed by passing the impure water through a filter paper. A mixture of sand and water can be separated by using filtration.

A circular filter paper is taken and folded twice to make a cone. This paper cone is kept inside a funnel. A beaker is kept below the funnel and the mixture is poured into the funnel. Sand is retained in the filter paper, and clean water is obtained in the beaker.



Filtration

Separation of Soluble Impurities

Evaporation

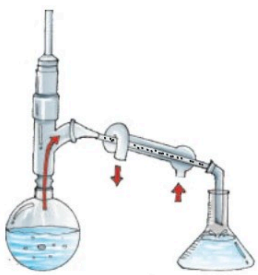
Some soluble impurities can be removed by heating the water. This process is called evaporation.

Just a Heads

Distilled water is used in vehicle batteries, medicines and for science experiments.

Distillation

It is a method which gives us both the solute and the solvent separately. On heating the solution, the water evaporates. It enters the condenser, which is kept cool by the cold water circulating around it. On entering the condenser, the hot steam changes into water again; which collects in the flask below. This water is totally pure and free from any impurities. It is called distilled water. The solute is left behind in the flask.


Distillation

Purifying Drinking Water

Water must be purified before drinking, as it may contain germs that can cause diseases. Water can be purified in the following ways.

- Water filters separate the impurities from water, leaving behind clean water.
- Potassium permanganate crystals can be added to wells, since these crystals kill germs.

Boiling

It is the simplest way to purify drinking water. Boiling water for about ten minutes kills the germs present in water. The boiled water should be stored in clean containers and kept covered.

Adding Chlorine

Chlorine is a chemical added to water to kill germs. This process of purifying water is called chlorination.

Make Teaching Fun

Encourage the students to come up with their ideas to state the importance of air and water. Discuss the composition of air by making a pie chart on the board. Ask them about the purification of drinking water. If possible organise a visit to the water works department. Tell them about your town's water supply.

Words Aid

- **Atmosphere:** the thick layer of air that surrounds the Earth
- **Decantation:** a water purification process in which water is slowly poured off into another container after sedimentation
- **Filtration:** the process of separating insoluble impurifies in liquid using a filter paper
- **Sedimentation:** the method of separating insoluble impurities in liquid by allowing them to settle down
- **Distillation:** process of separating liquids by heating and then collecting the vapour

Quick Review

- Our Earth is surrounded by a layer of gases called atmosphere.
- The atmosphere contains 78% nitrogen, 21% oxygen and 1% of other gases.
- Air has weight and occupies space. Air exerts pressure.
- Air is very useful to human beings and animals.
- Water covers about 78% of the surface of our Earth.
- Water is needed for drinking, cooking, washing clothes, bathing, etc.
- Drinking water can be purified by boiling, distillation or chlorination.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. The layer of gases that surrounds the Earth is called _____.
   a. mesosphere b. atmosphere
   c. distillation d. air pressure
2. Which layer protect us from the harmful ultra-violet rays of the Sun?
   a. Ozone b. Oxygen
   c. Carbon dioxide d. Nitrogen
3. It improves the rate of sedimentation _____.
   a. distillation b. heating
   c. alum d. filtering

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)
3. (a) (b) (c) (d)

4. The atmosphere contains 78% of _____.
   a. nitrogen b. carbon dioxide
   c. oxygen d. helium
5. A mixture of sand and _____ can be separated using a filter paper.
   a. sugar b. water
   c. salt d. air

4. (a) (b) (c) (d)
5. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

heat, humidity, troposphere, oxygen

1. We need _____ to breathe.
2. _____ is the nearest layer to the Earth's surface.
3. The amount of water vapour present in air is called _____.
4. _____ affects land and water differently.

C. Correct these statements by replacing the words underlined :

1. Air flowing to the land from the sea is called land breeze.
2. Hot air is heavier than cold air.
3. Water loses heat faster than the land.
4. Air exerts pressure in downward direction.

D. Define :

1. Atmosphere 2. Troposphere 3. Stratosphere
4. Humidity 3. Chlorination 6. Filtration

E. Answer the following questions :

1. Why is air important?
2. What is atmosphere? Why is it important?
3. Demonstrate the importance of air pressure with the help of an example.
4. What is sea breeze? With the help of a neat diagram explain it briefly.
5. Explain how insoluble impurities can be removed from water using a filter paper. Support your answer with the help of a diagram.
6. How can you make water safe for drinking in your house?

Sci-Fun

Give reasons, why?

1. Sound does not travel on the moon.

2. Mountaineers carry oxygen cylinders with them.
3. Water can enter a tumbler inverted in a bucket of water only when the tumbler is tilted slightly.
4. Water is used as a solvent to prepare many solutions.

Brain Teaser (HOTS)

- **The candle goes off when covered with a glass tumbler. Why?**
- **Will you get the same result if you use a piece of cloth instead of paper to prove that air occupies space?**

Subject Enrichment

Find the hidden words in the given word grid that you have learnt in this lesson.

S	E	D	I	M	E	N	T	S	W
A	T	M	O	S	P	H	E	R	E
W	S	S	X	O	U	E	N	A	A
A	A	A	Y	L	R	L	E	R	T
T	N	L	G	U	E	I	O	G	H
E	D	T	E	B	S	U	N	O	E
R	A	I	N	L	T	M	Y	N	R
W	I	N	D	E	A	R	T	H	Q

Enhance your Life Skills

- **By mistake your brother mixed some sand in a glass of salt water. What would you suggest him to separate them?**
- **We cannot live without air and water. These days we are facing problems due to increased level of pollution. What steps can you take to prevent water and air pollution?**

Subject Connect

Social Studies

- **Ozone is a natural gas which is present in atmosphere. Ozone layer present in stratosphere saves the environment by absorbing the harmful ultraviolet (UV) rays. Find out how ozone layer can be depleted and how we can prevent its depletion.**
- **What are the impacts of increased UV radiation on plants and aquatic ecosystems?**

9

Soil Erosion and Conservation

Intended Learning Outcomes

- Formation of Soil
- Soil Erosion
- Man's Activities
- Uses of Soil
- Agents of Soil Erosion
- Soil Conservation

Kick Off

The uppermost layer of the earth is called soil. It is made up of tiny pieces of rocks and remains of dead plants and animals. Soil contains many micro organisms.

Funtopia

What am I?

1. I make up the bottom layer of soil. _____
2. I am used for making earthen pots. _____
3. I am added to the soil to make it more fertile. _____
4. I am the hardest rock on earth. _____
5. I am found on the seashore and in desert. _____

Formation of Soil

The soil is the uppermost layer of the Earth's crust. It consists of pebbles of different size and remains of dead plants and animals. It also contains air and water.

Soil

Soil is very essential for all living things. It supports the growth of plants. Soil is also the home for many organisms like bacteria, fungi, worms and insects.

The process by which soil formation takes place is called weathering. Weathering is a very slow and gradual process during which parent rock breaks down into fine particles.

The quality of soil is an asset of a nation. It adds wealth to the country. The countries which are in deserts have to depend on other countries for food.

Uses of Soil

There are various uses of soil. All living beings depend on soil because their basic needs are fulfilled through the characteristics of the soil.

- Forests which grow on soil, bring rain and support various plant and animal activities.
- Microorganisms present in the soil make it fertile and help in maintaining the ecosystem.
- Large deposits of metals and minerals are present in the soil.
- Soil is also used for making bricks, pots and porcelain.
- Soil also stores water under the ground. This water is used for drinking and other purposes.

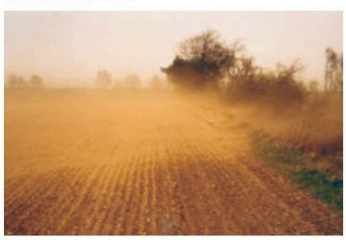
Activity

Place a sheet of paper on a table. Rub two rocks or big stones against each other, on the paper. You will see a kind of powder on the paper. This is how soil is formed when rocks rub against each other.

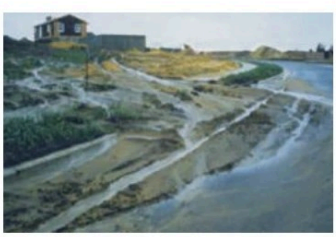
Soil Erosion

Natural forces like rain, wind and streams not only help in soil formation but they cause soil loss as well. As the wind blows and the water flows, they shift soil with them. This carrying away of soil by wind and water is called soil erosion.

Agents of Soil Erosion



Wind : Wind blows away the top layer of dry soil. The stronger the wind blows, the more soil it carries away. Erosion by wind reduces soil fertility and help in the spread of deserts. A dust storm is an example of soil erosion.



Water : River and rainwater cause soil erosion. When it rains, the topsoil from the hills come down to the rivers and plains. This running water deposits the mineral rich soil in river bed. During floods, the rivers get flooded and deposit the soil on another piece of land.

Activity

Take two boxes and fill them with soil. Plant some grass in one box. The other box must contain only soil. Keep both boxes tilted. Pour some water from the upper side on both boxes. You will find that the soil in the box without grass is carried away with water. But the soil in the box with grass is not washed away easily.

Man's Activities

Deforestation : The cutting down of trees in large numbers is called deforestation. It is the major cause of soil erosion. When the trees are cut down, soil becomes loose and is easily carried away.

Overgrazing : Overgrazing by animals in a particular area makes the land barren. This results in soil erosion.

Ploughing on hill sides : Ploughing of farmland also loosens the soil. Wind or water can easily carry this loosened soil away.

Deforestation

Overgrazing

Soil conservation means to prevent the soil from erosion. We must understand that nature took millions of years to create soil. If we do not preserve and protect it, then the natural balance will be disturbed badly. Soon there will be no life on earth.

We cannot stop natural forces like water, wind but we can take preventive measures to slow down the process of soil erosion.

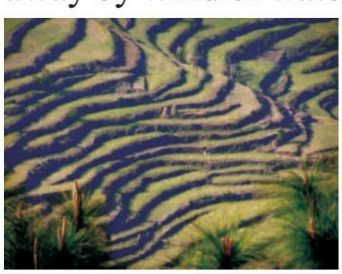
Soil Conservation

The different methods adopted to protect the soil from erosion are as follows :

Afforestation : Planting trees in large numbers is called afforestation. It is the most effective way of preventing soil erosion. Roots of plants hold the soil particles together and, thus, protect the soil from being carried away by wind or water.

Just a Heads up

Chambal valley in Madhya Pradesh is barren because of soil erosion by water.



Terrace Farming : Terrace farming is the step-like formation made on hill slopes to reduce the flow of water. The soil eroded from one step gets deposited on the next step and is never completely lost.

Fences : Farmers build fences around their fields. This prevents the rain water from washing away the topsoil and also break the speed of strong winds.

Embankments : Rivers constantly erode soil on their banks. Strong embankments of rocks and soil made along the banks prevent this erosion. High bunds can also prevent erosion of soil on nearby land by flooding.

Make Teaching Fun

Sunderlal Bahuguna, a noted environmentalist who initiated the Chipko Movement, was born on January 9, 1927. He had been fighting for the preservation of forests in the Himalayas.

The Chipko Movement was a forest conservation movement where people hugged on to trees to prevent them from being cut.

Words Aid

- **Weathering:** breaking of rocks by wind, water and microorganisms resulting, in the formation of soil
- **Soil erosion:** a removal of the fertile topsoil by wind, water, etc.
- **Soil conservation:** protection of soil against erosion

Quick Review

- Soil supports the growth of plants.
- Microorganisms present in soil make it fertile.
- Wind blows away the top layer of soil.
- The cutting down of trees in large numbers is called deforestation.
- Overgrazing by animals in a particular area makes the land barren.
- Terrace farming is the step-like formation made on hill slopes to reduce the flow of water.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. Which is the uppermost layer of the earth's crust?
   a. Minerals b. Soil
   c. Wind d. Water
2. The process of soil formation is called _____.
   a. afforestation b. deforestation
   c. weathering d. bedrock

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)

3. The carrying away of topsoil by natural agents is called _____.
   a. subsoil b. soil erosion
   c. soil conservation d. weathering
4. Soil erosion can be controlled by _____.
   a. overgrazing b. conservation
   c. deforestation d. afforestation

3. (a) (b) (c) (d)
4. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box:

metals, fences, soil, minerals, deforestation, soil erosion

1. The quality of _____ is an asset of a nation.
2. Large deposits of _____ and _____ are present in the soil.
3. River and rainwater cause _____ .
4. _____ is the major cause of soil erosion.
5. Farmers build _____ around their fields.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Soil does not support the growth of plants. ☐
2. Afforestation causes soil erosion. ☐
3. Building embankments saves the soil. ☐
4. Water causes soil erosion. ☐
5. Microorgansms present in the soil make it infertile. ☐

D. Write one word for the following :

1. The topmost layer of the earth _____
2. Cutting down of trees in large scale _____
3. Farming done on the steps of hill slopes. _____
4. Protecting soil from erosion. _____
5. Structures built along the banks of nearby fields. _____

E. Answer the following questions :

1. Explain the process of soil formation.
2. What are the uses of soil?
3. What is soil erosion? What are the agents of soil erosion?
4. What do you understand by soil conservation?
5. What is the difference between afforestation and deforestation?
6. What is terrace farming?

Sci-Fun

Find the words hidden in the given word grid.

E	T	E	R	R	A	C	E	F	H
R	A	I	N	F	A	L	L	A	I
O	Q	C	O	N	T	O	U	R	L
S	O	I	L	C	R	O	P	M	L
I	D	T	A	W	I	N	D	I	D
O	A	O	N	S	E	E	D	N	R
N	M	P	D	R	O	C	K	G	Y
V	E	G	E	T	A	T	I	O	N

Brain Teaser (HOTS)

- **Terrace farming is also known as step farming. Why?**
- **Sohan lives in Darjeeling. He does farming there. What method should he use to prevent soil being eroded?**

Subject Enrichment

Divide the class into four groups and give them the following topics for discussion. At the end of discussion, the groups come back and present their points in front of the class.

a. Soil conservation
b. Ill effects of deforestion
c. Layer of soil
d. Types of soil

Enhance your Life Skills

- **On the occasion of World Environment Day, your friends and you decided to plant trees in the school garden. Your teacher said that trees would also help in soil conservation. Why do you think the teacher said so?**
- **Why is soil considered the foundation of life?**

Subject Connect

Art and Craft

Use clayey soil to make small toy animals or some other objects. Keep them in the Sun to dry. You can paint them once they are dry and hardened.

10 Rocks and Minerals

Intended Learning Outcomes

- Rocks
- Sedimentary Rocks
- Metamorphic Rocks
- Uses of Rocks and Minerals
- Coal, Petroleum
- Types of Rocks
- Igneous Rocks
- Minerals
- Fossil Fuels
- Conservation of Fossil Fuels

Kick Off

Rocks are all around us. You can see rocks inside your house, in your yard, in stores, everywhere you look. Statues, chalk, marble, pencils, mountains, pebbles, soil, and volcanoes are all made of rocks, no matter where you are. You are always close to rocks and minerals.

Funtopia

Identify the pictures and solve the riddle.

a. A beautiful building in India is made out of me.

I am made of M _ _ _ _ _.

b. I am a soft rock. You can use me to write on a blackboard.

I am a C _ _ _ _.

Rocks

You have read that soil is formed by the weathering of rocks. The Earth is made up of rocks. Have you ever wondered how rocks are formed?

All rocks are made of substances called minerals. Minerals are the building blocks of Earth's rocks. The rocks are different due to the presence of different minerals in different quantities. A mineral is a non-living natural substance which is neither plant nor animal.

Both rocks and minerals are very useful for our daily life.

The study of rocks and minerals is called geology, and the scientists who study these rocks and minerals are called geologist.

Types of Rocks

Many of the materials that we use everyday are made from different kinds of rocks. Each kind of rock is made of different minerals. Depending on their composition, formation and degree of hardness rocks are classified as :

i. Sedimentary rocks ii. Igneous rocks and
iii. Metamorphic rocks

Sedimentary Rocks

A river carries, or transports, pieces of broken rock as it flows along its course. When the river reaches a lake or the sea, its load of transported rocks settles to the bottom. The deposited rocks build up in layers, called sediments. This process is called sedimentation.

The weight of the sediments on top squashes the sediments at the bottom.

This is called compaction. The water is squeezed out from between the pieces of rocks and crystals of different salts are formed.

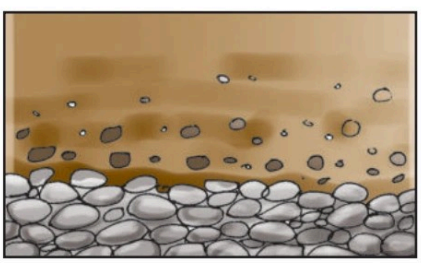
The crystals form a sort of glue that sticks or cements the pieces of rocks together. This process is called cementation.

These processes eventually make a type of rocks called sedimentary rocks. It may take millions of years for sedimentary rocks to form.

These are the different processes in order :

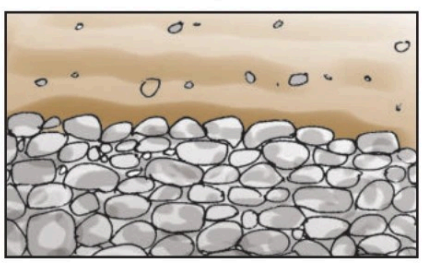
Sedimentation ⟶ Compaction ⟶ Cementation

Sedimentation



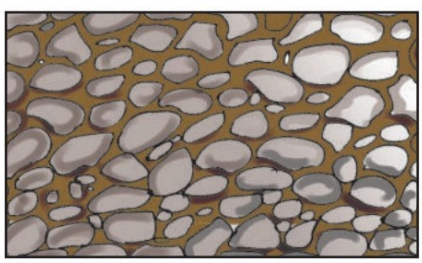
The river transports bits of rock and deposits them on the bottom of the riverbed.

Compaction



Deposited rocks build up in layers, and the weight of the layers compresses the bottom layers.

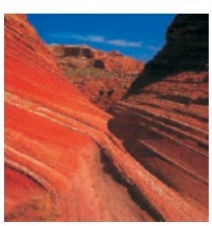
Cementation



The compression squeezes out water, leaving salt crystals that cement the rocks together.

Sedimentary rocks contain rounded grains in layers. Examples of sedimentary rocks are :

- Chalk
- Limestone
- Sandstone
- Shale



Sedimentary rocks like sandstone have layers

The oldest layers are at the bottom and the youngest layers are at the top.

Sedimentary rocks may contain fossils of animals and plants trapped in the sediments as the rocks were formed.

Igneous Rocks

The second type of rock we'll look at is igneous rock.

Formation

The inside of the Earth is very hot; hot enough to melt rocks. Igneous rocks form during volcanic eruptions. Hot molten rock flows out of volcano; cools and solidifies to form igneous rocks.

What are they like?

Igneous rocks contain randomly arranged interlocking crystals. The size of the crystals depends on how quickly the molten magma solidified. The more slowly the magma cools, the bigger the crystals.

Almost 95 per cent of the Earth's crust is made up of igneous rocks. These contain minerals like mica, feldspar, and magnesium.

Granite : It is very hard igneous rock that has been formed by the slow cooling of lava. It is used extensively as a building material. You might have seen polished granite on kitchen counters, floors, etc. It comes in many colours.

Granite

Obsidian

Obsidian : It forms by quick cooling of lava. It is smooth and glassy. It is usually black or dark-coloured. Obsidian is used in making jewellery and ornaments.

Pumice : It is a light ash-coloured rock formed by quick cooling of lava. It has many holes in it formed by the gases trapped in lava while it was cooling. It resembles a sponge and is very light weighted. It has long been used as an abrasive in cleaning and polishing. It is also used to make poured concrete, tile, and plaster. Feldspar and zircon are some minerals found in pumice.

Pumice

Unlike sedimentary rocks, igneous rocks do not contain any fossils. This is because any fossils in the original rock will have melted when the magma was formed.

Metamorphic Rocks

Metamorphic rocks are rocks that have changed (to metamorphose is to change form). They may be formed due to physical and chemical changes in igneous, sedimentary, or older metamorphic rocks themselves. These change happen due to heat and pressure.

Marble : Marble is an example of a changed sedimentary rock. It is formed from limestone. Marble is used in making statues and for other ornamental purposes. It is quite popular as a material for flooring too.

Marble

Slate

Slate : Slate is also metamorphosed sedimentary rocks, formed from shale. Like shale, it splits into flat layers. It is used in making bricks and cement.

Gneiss : Gneiss (pronounced as 'nice') is formed from granite. It has different kinds of minerals in it, giving it a striped or streaked look. It is used as building stone and for making other structures.

Gneiss

Quartzite

Quartzite : It is formed from sandstone that is rich in the mineral quartz. It is a very hard rock and is weather-resistant. It is used in the glass and ceramic industry and also in jewellery and ornaments.

Minerals

Minerals are natural substances made of chemicals, either a single chemical by nature or a combination of chemicals. They are inorganic, which means that they are not made by any living organism.

Uses of Rocks and minerals

Rocks and minerals are all around us. They are a vital part of our everyday lives. They provide us with shelter and even food, as well as innumerable objects that make our existence easier or more enjoyable. They make up our television, cars and clothing, computers and even our currency.

- Rocks are used as building material—to make bricks, cement, tar etc.
- Rocks give us useful metals—Metals like iron and copper which are used to make household articles are made from ores obtained from rocks. Precious metals like gold and silver are used to make ornaments.

- Some minerals like uranium are used to produce tremendous energy, known as nuclear energy.
- Some rocks are precious and are used as gemstones—Diamonds, rubies and emeralds are some gemstone. When polished, they are used in jewellery.

Bricks and building materials, gold and silver, diamond and other stones

Some rocks give us fuel—Coal is one of the most important sources of fuel. It is used for heating, electricity production, for moving engines, etc. Petroleum (obtained from dead remains of animals) is very valuable and is called black gold. It gives us many different kinds of fuels like LPG, CNG, diesel, petrol, and kerosene. Petroleum is also used in medicines, polish making, printing, inks, dry cleaning, and so on.

Humans need to consume a variety of minerals to maintain good health. These minerals support and strengthen the body's different systems, including the skeleton, bloodstream, and nervous system. For example : Calcium is essential for building strong bones. Other minerals just make food taste good, such as salt.

Fossil Fuels

Fossil fuels are natural substances made deep within the Earth from the remains of ancient plants and animals. Over time, heat and pressure turn decomposing remains into fuels, which release energy on burning. Coal, oil and natural gas are the three main fossil fuels. They were made underground in similar ways. The type of fossil fuels created depends on the kinds of plants or animals they are made from, and the amount of heat and pressure.

Coal

Coal was developed from the remains of plants that got buried deep inside the Earth millions of years ago.

Over the years, these plant remains which were exposed to heat and tremendous pressure changed to coal.

In steel industry

In cooking

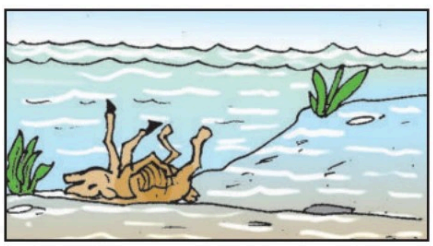
In steam engines

Uses of coal

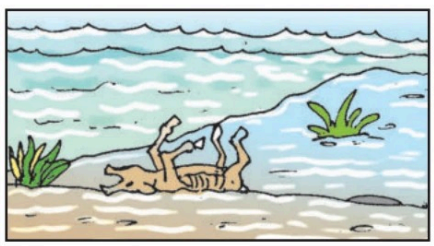
Coal is found deep under the ground. It is used for making steel, for cooking and also in the steam engines. It is also used to produce electricity and many substances such as plastic and nylon.

Petroleum

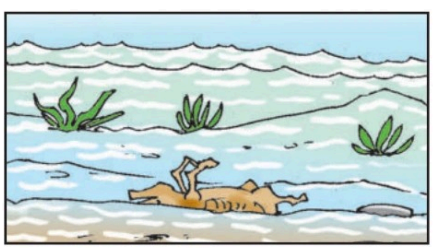
Petroleum is another underground resource. Petroleum is formed from remains of plants and animals that lived in the sea millions of years ago. These creatures got buried deep under the bottom of the sea and gradually changed into petroleum.



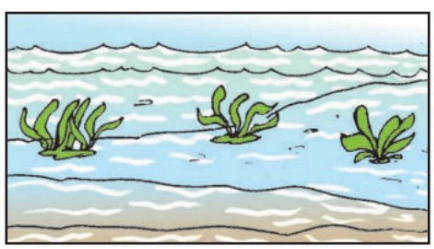
Sea plants and animals died and are buried on ocean floor.



Over time, they are covered by layers of silt and sand.



Over million of years the remains are buried deeper and deeper.



The enormous pressure and heat turn them into soil and gas.

Formation of Petroleum

Just a Heads up

There are three kinds of coal Lignite, is buried just below the surface. It was formed more recently and give less heat. **Bituminous** and **anthracite** coals both are black and found far below the ground. Anthracite coal produces the most heat and Bituminous coal produces heat but less than anthracite.

Petroleum products

Wells are dug deep under the sea to bring this petroleum to the surface. It is heated in refineries and changed into petrol and diesel for vehicles. It is used for dry cleaning woollen clothes and also used for making shoe polishes, paints and varnish. It is used as a lubricant for machines. Some useful petroleum products like cosmetics, medicinal ointments, paraffin wax, etc., are used by us daily.

Conservation of Fossil Fuels

Deposits of fossil fuels such as coal, gas and oil are limited. They took millions of years to be formed. This is why we should be careful not to waste them. If we use them carefully, they will last longer.

Fossil fuels take millions of years to form through gradual natural processes. They will not last forever. If the deposits of fossil fuels are exhausted, they will no longer be available to us. Also, burning of

Just a Heads up

Geothermal energy is heat from within the Earth.

fossil fuels cause pollution. So, we need to conserve fossil fuels and use alternative sources of energy. Such as solar energy, hydro-power and wind energy that are always available to us and are non-polluting.

Solar panels

A solar cooker is a device which uses solar energy.

Make Teaching Fun

Take playing clay of three to four different colours. MIx them together to form one solid ball-like structure. Let it dry and become hard. You will see that the ball is made up of different colours. This is like a rock with different minerals.

Words Aid

- **Rocks:** solid and usually materials of which the Earth is made of
- **Minerals:** materials of which rocks are made
- **Igneous rocks:** rocks formed due to cooling of lava
- **Sedimentary rocks:** rocks formed by continuous deposition of small particles or sediments exposed to continuous pressure.
- **Metamorphic rocks:** rocks formed due to changes in igneous or sedimentary rocks

Quick Review

- Rock is a naturally occurring aggregate of minerals.
- Igneous, Sedimentary and Metamorphic rocks are three kinds of rocks.
- Igneous rocks such as granite and basalt are formed when magma cools, either below or above the surface of the Earth.
- Sedimentary rocks are formed by the accumulation of sediments on the beds of rivers and oceans.
- Metamorphic rocks are formed from the sedimentary and igneous rocks under the effect of heat and pressure.
- The inorganic substances found in the rocks in the form of crystals are called minerals.
- Coal is a sedimentary rock formed from the remains of plants.
- Petroleum is formed from the remains of plants and animals in the sea.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. Which of the following is the order for forming sedimentary rocks?
   a. Sedimentary - cementation - compaction
   b. Compaction - sedimentation - cementation
   c. Sedimentation - compaction - cementation
   d. Cementation - Sedimentation - Compaction
2. Only one of these rocks is a sedimentary rock. Which one?
   a. Shale b. Slate
   c. Granite d. Marble
3. Where are the oldest layers of rocks usually found in a cliff made from sedimentary rocks?
   a. At the top b. In the middle
   c. At the bottom d. Every where
4. What is magma?
   a. Salt crystals in sedimentary rock b. Heater crystals
   c. Bubbles of gas d. Molten rock
5. Which type of rock never contains fossils?
   a. Sedimentary b. Igneous
   c. Metamorphic d. Both a. and c.
6. What does the presence of tiny crystals in a piece of igneous rock tell you about it?
   a. The molten rock cooled very quickly
   b. The molten rock cooled very slowly
   c. The molten rock cooled deep underground
   d. The molten rock remained so
7. Which of these rocks is an igneous rock?
   a. Marble b. Limestone
   c. Basalt d. Slate
8. Which statement about metamorphic rocks is correct?
   a. They are formed when rocks are heated until they melt.
   b. They are only formed from heated sedimentary rocks.
   c. They are formed from all types of rock.
   d. They may contain fossils of animals

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1.	a	b	c	d
2.	a	b	c	d
3.	a	b	c	d
4.	a	b	c	d
5.	a	b	c	d
6.	a	b	c	d
7.	a	b	c	d
8.	a	b	c	d

9. Which statement about the crystals in metamorphic rocks is correct?
   a. They are small if the molten rock cooled quickly.
   b. They are large if the molten rock cooled quickly.
   c. They are often arranged in layers.
   d. They are formed in a short period of time.
10. Which metamorphic rock is formed from limestone?
   a. Marble
   b. Slate
   c. Shale
   d. Quartzite

9. (a) (b) (c) (d)
10. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

igneous, petroleum, minerals, minerals, fossils, pumice, uranium, rocks, nuclear

1. All rocks are made of substances called _____.
2. The study of _____ and _____ is called geology.
3. _____ rocks form during volcanic eruptions.
4. _____ is the only rock that can float.
5. Minerals like _____ are used to produce tremendous energy, also known as _____ energy.
6. Igneous rocks do not contain any _____.
7. _____ is formed from remains of plants and animals which lived in the sea.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Sedimentary rocks are formed from igneous rocks.
2. Granite is a metamorphic rock.
3. Coal and petroleum are fossil fuels.
4. Rocks are of three kinds depending on how they are formed.
5. The deposits of coal and petroleum are formed very fast.
6. Coal is an igneous rock formed from the remains of animals.

D. Define the following :

1. Geologist
2. Sedimentation
3. Geology
4. Solar Cooker

E. Answer the following questions :

1. Define rocks and minerals.
2. Differentiate between igneous and metamorphic rocks.

3. Write the various uses of rocks and minerals.
4. How are fossils formed?
5. Describe the formation of coal and petroleum.

Sci-Fun

Read the clues and write the names of minerals below each picture with the help of the given words.

chalk gold granite

1. It is used to make kitchen slabs.

_____

2. It is used to write on the blackboard.

_____

3. It is used for making jewellery.

_____

Brain Teaser (HOTS)

- **Some foot scrubbers that are made from an igneous rock called pumice can float on water. Why?**
- **Design a house for yourself. Give details of the types of rocks you would use in the construction of various parts—floor, walls, etc. and why?**

Subject Enrichment

Have you ever seen stones used to make a heel-scrubber or (a) mortar and pestle? Find out, which stones they are made of.

Enhance your Life Skills

- **Make a list of the types of rocks used in your school building.**
- **Edible rocks: use wafers, chocolate, cake slices, biscuits, jell, etc; to create a model of sedimentary rocks.**
- **Use plaster of paris to show the process of formation of igneous rocks.**

Subject Connect

Geography

Take a map of India and mark the following regions:

1. A place where granite is extracted.
2. A state that has reserves in marble.
3. A place where petroleum is extracted.

11 The Earth and The Solar System

Intended Learning Outcomes

- The Earth
- The Sun
- The Moon
- The Surface of the Moon
- Artificial Satellites
- Humans on the Moon
- Shadows
- The Tides

Kick Off

In a previous grade, we have learned about solar system that consists of the Sun, eight planets, their satellites and other heavenly bodies like asteroids, comets.

Funtopia

Read the following and solve the riddle.

1. I am known as the red planet. I have two moons. What am I? _____
2. I am the outermost layer of Earth. I mostly consist of rocks such as granite and basalt. I am the _____.
3. I am the second largest planet in the solar system. My Hindi name is _____.
4. I take approximately 12 years to complete one revolution. I spin faster than any other planet. How many moons do I have? _____
5. I am an opening on the surface of the Earth through which hot gases and molten rocks escape. I am a _____.

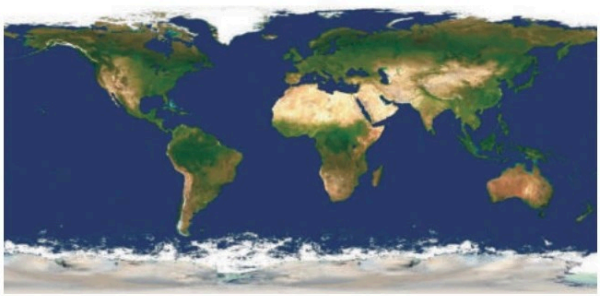
Do you remember that the Earth is a part of the solar system and that it revolves around the Sun. Earth is the only known planet to support life and the Moon is its natural satellite and revolves around the Earth. All the planets, Moon and the stars are called celestial bodies. The science that deals with the study of all fascinating celestial bodies is called astronomy.

People who study about heavenly bodies are called astronomers. People who travel in a spacecraft are called astronauts. Kalpana Chawla was the first Indian-born woman astronaut.

The Earth

The Earth is different from all other planets in the solar system. We have vast quantities of water on our planet, and we have oxygen to breathe. We have soil in which plants grow. There is a blanket of air around the Earth, called the atmosphere, that protects it from the harmful ultraviolet rays of the Sun.

We know that the Earth is like a big ball covered by water and rock. The surface of the Earth has various features such as oceans, rivers, mountains, and valleys. But do you know what the inside of the Earth looks like?



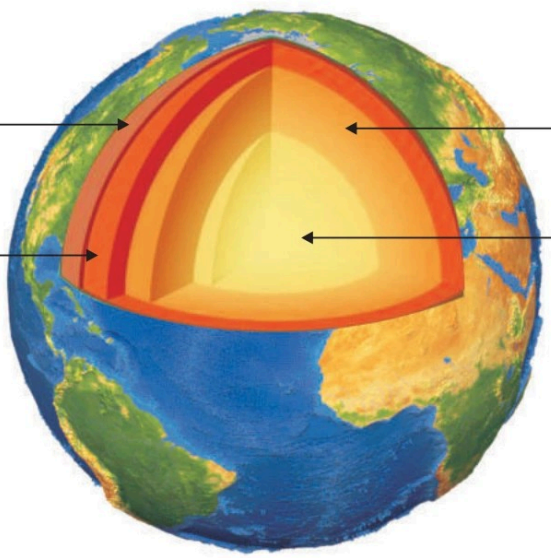
Surface of the Earth

Internal Structure of the Earth

According to scientists, the Earth is not a uniform solid sphere. It consists of different layers. These layers are called the crust, mantle and core.

The **crust** varies in thickness from 5 km under the ocean floor to 70 km under the continents.

The **mantle** is about 2900 km thick. Its upper part is made of solid rock, whereas the lower part consists of molten rock. The rocks in the mantle are mainly composed of iron and magnesium.



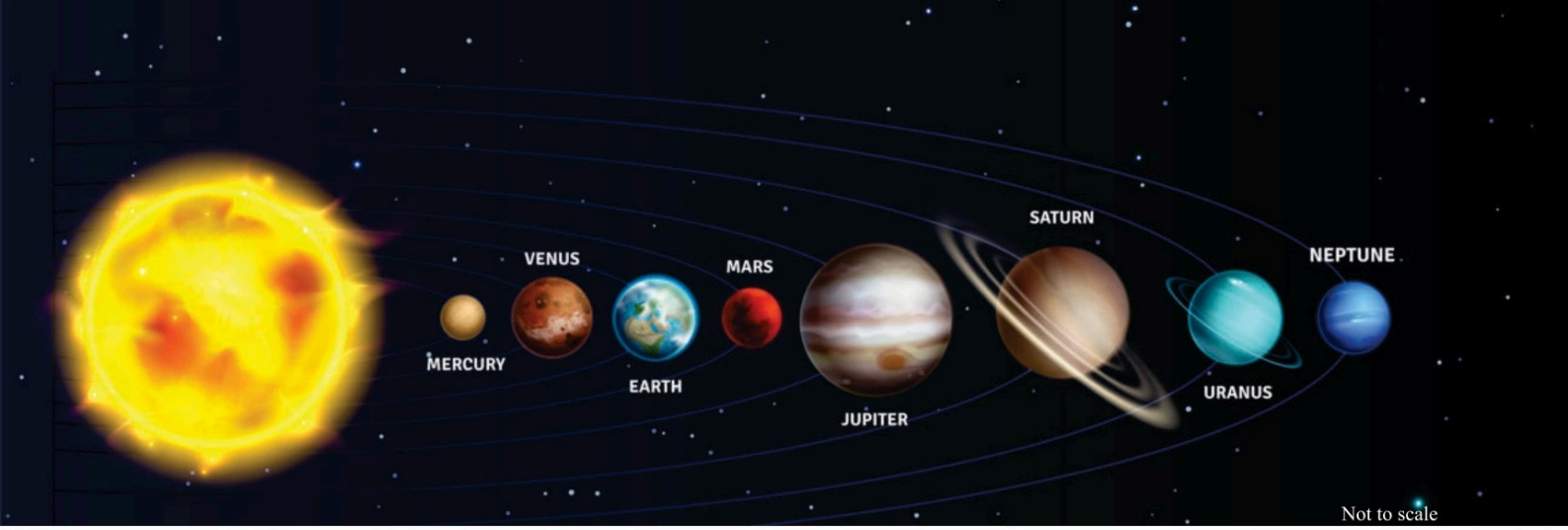
The **outer** core is the only liquid layer of the Earth. It is about 2300 km thick. It contains iron and nickel in a molten state. There is also some sulphur present here.

The **inner core** is an extremely hot, solid sphere made mostly of iron and nickle at the centre of the Earth.

Internal Structure of the Earth

The Sun and the celestial bodies that orbit it form the solar system. The Sun is the centre of the solar system. All the planets revolve around the Sun in fixed paths called the orbit. Orbits of the planets are elliptical (oval in shape). Planets are heavenly bodies that revolve around the Sun.

The planets of the solar system in order of increasing distance from the Sun are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.



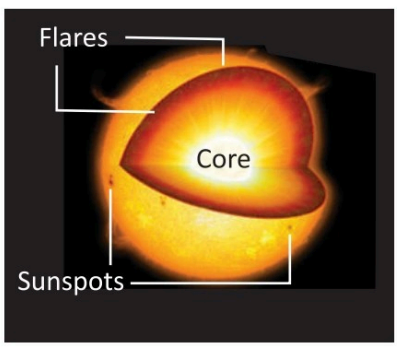
The Solar System

Pluto, which was earlier considered a planet, was placed in a separate category called 'dwarf planets' by the International Astronomical Union (IAU) in August 2006.

The Sun

The Sun is the star at the centre of the solar system. It is a hot ball of gases that gives off great amounts of energy. Life on Earth depends on light and heat from the Sun.

All the heavenly bodies present in our solar system orbit or travel around the Sun. Earth orbits the Sun at a distance of about 15,00,00,000 kilometres. The next closest star is much, much farther away—about 2,50,000 times farther than the Sun. That is why the Sun is by far the brightest object in the sky.



The Sun

The Sun is the largest object in our solar system. It has a diameter of about 13,92,000 kilometre. This is about 109 times bigger than Earth's diameter.

The Sun consists mostly of the gases hydrogen and helium. The hydrogen is packed in tightly at the core. Great pressure squeezes the hydrogen, causing it to change into helium. Its core is very hot, reaching about 15,600,000°C.

The Sun has been shining for millions of years. It is now mostly hydrogen with some helium. Billions of years in the future, the Sun will use up all its hydrogen fuel. Eventually it will stop shining.

The surface of the Sun is made of gases. The surface temperature is about 6,000°C. Sometimes cooler, darker patches called sunspots appear on the surface.

Sometimes great blasts of energy, called solar flares, erupt from the atmosphere of the Sun.

The Moon

The Moon is also a ball of rocks like the Sun and the planets. It doesn't give out light of its own. It reflects the light of the Sun. The Moon is the natural satellite of the Earth. Its distance from the Earth is 384,400 km.

The Surface of the Moon

The surface of the Moon is covered with a layer of dust. It has mountains, craters and valleys. Craters are deep holes that have been made when solid bodies called meteorites crashed into the moon's surface. Lava that flowed from these craters cover the earlier craters and they look smooth and dark now.

There is no life on the Moon because it has no air or water. There is no atmosphere surrounding the moon. So there are no winds, no clouds and no rain. On the Earth the atmosphere absorbs a part of sunlight, so it does not become so hot during the day. But there is no protection against the strong rays of the Sun on the Moon. The side facing the Sun is very hot and the side facing away from it is very cold. If you shout on the Moon, nobody can hear you! This is because sounds travels in air, and there is no air on the Moon.

Just a Heads up

The Earth pulls everything down with the force of **gravity**. That is why anything thrown up, comes down. The Moon also pulls all things towards itself. But since it is much smaller than the Earth, the force of gravity is less. It is only about one-sixth of that on the Earth. This means that if you can jump 1 metre high on the Earth, you can jump 6 metres high on the Moon with the same effort. In fact your weight on the Moon would only be one-sixth of your weight on the Earth!

Artificial Satellites

The Moon is the only natural satellite of the Earth. A heavenly body which revolves around a planet is called a satellite. Natural satellites of planets are known as Moons. The Earth has only one moon.

Artificial Satellite

The Moon is the only natural satellite of the Earth. Artificial satellites are man-made objects sent into space to orbit planets. The first man-made satellite was Sputnik-I which was sent into space on 4th October, 1957. Since then, many countries have sent their satellites into space. *Aryabhatta* was India's first satellite.

Artificial satellites are extremely useful to us. Some of them are :

- **Communication satellites**—like INSAT-3E send telephone calls and TV programmes to different parts of the world.
- **Weather satellites**—help predict the weather. You may have seen satellite pictures on TV taken from space. These satellites give early warnings of storms and cyclones.
- **Remote sensing satellites**—take photographs of the Earth. Scientists use these to study the features of the Earth's surface. Cartosat-1 is one such satellite from remote sensing satellite launched by India.
- **Science satellites**—help study space and galaxies far away.

Just a Heads up

Artificial satellites are launched into space by rockets. Once beyond the Earth's gravity and atmosphere, they are sent circling around the Earth by the force of another rocket. There is nothing in space to interfere with the path or speed of the satellites, so they keep orbiting the Earth.

Humans on the Moon

Three American astronauts—Neil Armstrong, Edwin Aldrin, and Michael Collins—went to the Moon in a spacecraft called Apollo 11. On 21 July 1969, Neil Armstrong and Edwin became the first two people to walk on the Moon.

Shadows

Shadows are formed when an opaque body obstructs the path of light. Shadow can never be formed in total darkness.

Opaque objects are objects through which light cannot pass, they form clear and dark shadow, e.g. wood and metal are opaque.

Transparent objects are objects through which light can pass totally. They do not form shadow, e.g. plain glass or clean water are transparent.

Translucent objects are objects through which light can pass partially. They form faint shadow e.g. butterpaper and a clean plastic bags are translucent.

Opaque

Transparent

Translucent

Twisting Time

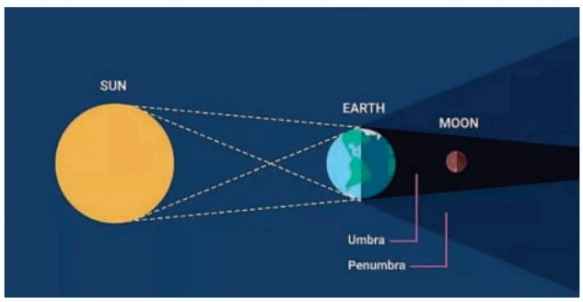
Make a list of 10 objects you see around you. Classify them as opaque, transparent and translucent.

Just a Heads up

Heavenly bodies other than the Earth and the moon can also eclipse each other. The planet Jupiter sometimes blocks light from its moons.

Eclipse : An eclipse is a shadow formed in space which makes the Sun or the moon invisible for some time.

Lunar eclipse : It occurs on a full moon night. As the moon revolves around the Earth, the Earth's shadow falls on it. So we cannot see the moon. The Earth comes between the Sun and the moon. If the complete disc of the moon is covered by the shadow of the Earth, it is known as a total lunar eclipse. If only a part of the moon is hidden, it is called a partial lunar eclipse.

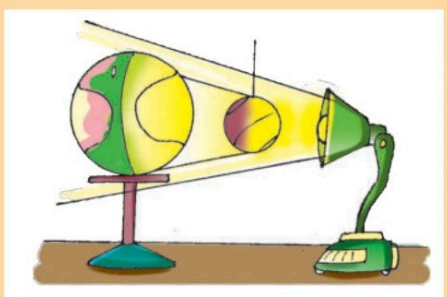


Lunar Eclipse

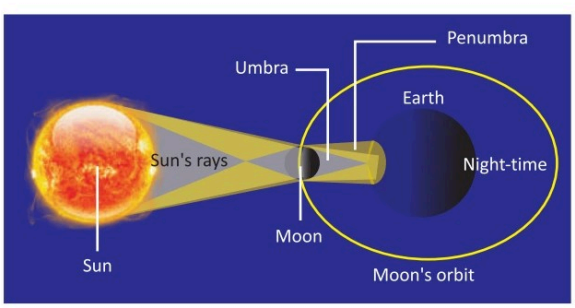
Activity

To study a lunar eclipse.

Take a basketball and place it on a stand on a table. This represents the Earth. Tie a string in the form of a sling around a tennis ball and hold it on the left of the basketball and place a powerful table lamp to the right of the basketball. The table lamp represents the Sun, and the tennis ball, the moon. Now move the 'moon' around the 'Earth'. This will be the position in which a lunar eclipse will occur.



Solar eclipse : An eclipse of the Sun is called a solar eclipse. It occurs on a 'new moon' day. It occurs when the moon comes between the Sun and Earth. The moon casts its shadow on the Earth and people in certain parts of the Earth cannot see the Sun at all or can only see it partially.



Solar Eclipse

For a total solar or lunar eclipse, the Sun, the Earth, and the Moon have to be in a straight line.

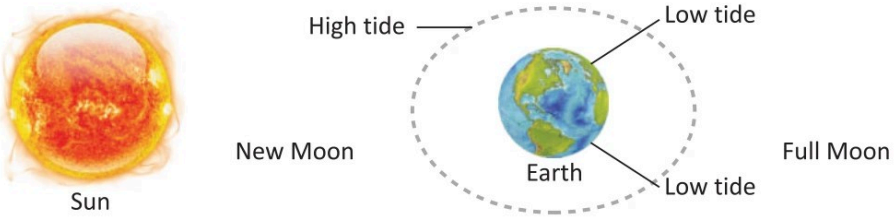
Activity

Set up the experiment as done for a lunar eclipse, but the tennis ball (moon) should be between the lamp (Sun) and the basketball (Earth). Move the 'moon' around the 'Earth' till it comes to lie in such a position that it casts a shadow on the 'Earth'. This will be the position in which a solar eclipse occurs.

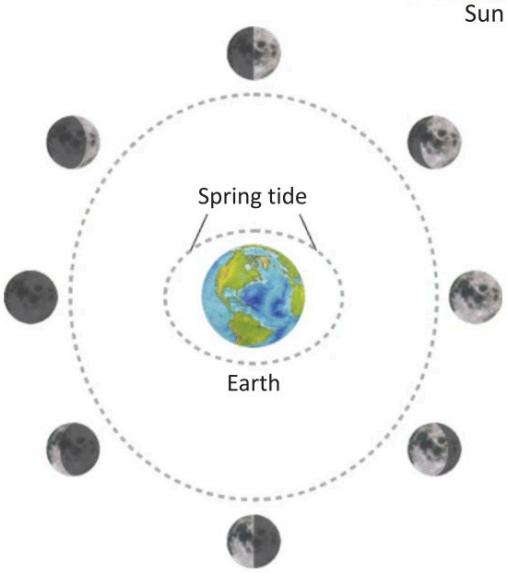
The Tides

The regular rise and fall of the level of water in the sea is called a tide. Tides are related to the Sun and the moon. The moon is much closer to the Earth than the Sun. It, therefore, has greater pull on water than the Sun. The moon's gravitational pull attracts the water of sea. This causes the water to rise towards the moon. This is called high tide. At high tide, the water in the sea rises on the sea shore. Another high tide occurs on the opposite side of the Earth, because of the Earth's movement. In the region between the two high tides, the level of water falls and forms low tide. Thus, a high tide and a low tide occur almost every 12 hours.

The knowledge of tides is useful to ships and fishermen. When there is high tide, ships and fishermen can come easily to a port and in low tide, they cannot bring back their ships to the shore because the sea water recedes far away from the shore.



High and low tides



Spring tides

When the moon is full or new, the gravitational pull of the moon and the Sun are combined. At these times, the high tides are very high and the low tides are very low. This is known as a spring high tide. Spring tides (they do not have anything to do with the season spring) occur when the Earth, the Sun and the moon are in a line. The gravitational forces of the moon and the Sun both contribute to the tides. Spring tides occur during the full moon and the new moon.

Make Teaching Fun

Chandrayaan-1 was India's first lunar probe. It was launched by the Indian Space Research Organisation (ISRO) in 2008. The mission of the probe was to explore the moon. The probe operated until late-2009. Now, collect information about Chandrayan-3.

Words Aid

- **Artificial Satellite:** a man-made object that orbits the earth or any other planet
- **Astronaut:** a person who is trained to travel in a space craft
- **Eclipses:** shadows formed when the Sun, Earth and Moon come in a straight line, with the Sun at one end
- **Space shuttle:** a rocket launched in space to make repeated journeys between Earth and space
- **Tides:** rise and fall of water on the Earth due to gravitational pull of the moon and the Sun
- **Satellite:** an object revolving around a planet
- **Crust:** the outermost layer of the Earth
- **Mantle:** the layer of the Earth between crust and the core
- **Core:** the innermost layer of the Earth

Quick Review

- All the planets, satellites and stars are called celestial bodies.
- The Sun is the nearest star to the Earth.
- Earth is like a big ball covered with water and rocks.
- The Earth consists of three layers : crust, mantle and core.
- The moon is our nearest neighbour in space.
- The surface of the moon is covered with craters, big mountains and valleys.
- There is no life on the moon because there is no water and air.
- A solar eclipse occurs when the shadow of the moon falls on the Earth.
- A lunar eclipse occurs when the shadow of the Earth falls on the moon.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. All planets revolve around the _____.
   a. Sun b. Earth
   c. Mars d. Moon
2. The inner part of the core of the Earth is _____.
   a. woody b. liquid
   c. gas d. solid
3. The deep holes on the surface of the moon are called _____.
   a. craters b. cores
   c. meteors d. eclipse

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)
3. (a) (b) (c) (d)

4. The gravity of the moon is _____ of the gravity of the Earth.
   a. sixth b. one-sixth
   c. one-fifth d. one-third
5. The regular rise and fall of the level of water in the sea is called a _____.
   a. storm b. lunar
   c. tide d. satellite
6. The spacecraft which successfully landed on the moon was _____.
   a. Apollo 11 b. INSAT 3E
   c. Cartosat-1 d. Chandrayaan-1

4. (a) (b) (c) (d)
5. (a) (b) (c) (d)
6. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

Solar System, Moons, Spring, Earth, tides, astronomers, International Astronomical Union

1. _____ is the only planet which supports life.
2. People who study about heavenly bodies are called _____.
3. The Sun and the celestial bodies that orbit it form the _____.
4. IAU stands for _____.
5. _____ of planets are known as natural satellites.
6. The knowledge of _____ is useful to fishermen.
7. _____ tides occur during the full moon and the new moon.

C. Differentiate between :

1. Solar and Lunar eclipse
2. Transparent and Opaque objects
3. Natural and Artificial satellites
4. Satellites and Moons

D. Write one word for the following :

1. Blanket of air around the Earth _____
2. The Sun and the celestial bodies that orbit it _____
3. Heavenly bodies that revolve around the Sun _____
4. The dwarf planet _____
5. The natural satellite of the Earth _____

E. Give reasons, why?

1. If you shout on the moon no body can hear you.
2. You weigh less on the Moon.

3. When a big rock crashes on the surface of the Moon, it cannot be heard.
4. If we throw a ball up it comes back to the Earth.

F. Answer these questions :

1. Why is life possible on Earth?
2. Explain the internal structure of the Earth. Draw the related diagram.
3. Why is there no life on the Moon?
4. Explain how tides occur.
5. Write a few lines about the first visit to the Moon.
6. How are weather satellites useful?

Sci-Fun

- **Eclipse is a kind of shadow. There are two types of eclipses—Lunar eclipse and solar eclipse.**
- **Collect interesting facts about the surface of the moon. Also, collect information about the various exploration missions to the moon. Prepare a chart and present the information you have collected to the class.**

Brain Teaser (HOTS)

- **Imagine that you are going to the outer space. What all things would you carry with you? Make a list in your science notebook.**
- **What would happen when all the hydrogen in the Sun is converted into helium?**

Subject Enrichment

- **Movement of the moon around the Earth gets completed in 27.3 days. We assume this period as a month. That's why 12 months are given in the calendar showing 12 revolutions of moon around the Earth.**

Enhance your Life Skills

- **Tides are affected by the phases of the moon. The highest tides are seen on new moon days and lowest tides are seen on the full moon days. Knowledge of tides helps people to know when to go out near the sea.**
- **Why is the Moon so important to life?**

Subject Connect

Language

Search for poems written about the Moon in English, Hindi or any other language that you speak at home. Recite them in class and discuss their meanings.

12 Natural Disasters

Intended Learning Outcomes

- Earthquakes
- Tidal Waves and Tsunami
- Volcanoes
- Floods
- Droughts

Kick Off

A natural disaster is a terrible event that is caused by the forces of nature and bring with it a lot of destruction, damage, pain and suffering. Let us take a close look at some natural disasters.

Funtopia

Look at the pictures given below and identify the natural disasters. Choose the words from those given below.

Earthquake Drought Flood Volcanic eruption

_____

_____

_____

_____

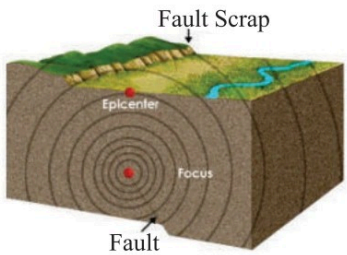
Earthquakes

An earthquake is a sudden shaking or tremor of the Earth caused due to movements or vibrations inside the Earth. Let us know how earthquakes occur.

As we know, the Earth is made up of three layers—crust, mantle and core. The outermost layer of Earth is called the crust. The layer below the crust, known as the mantle, is made up of hot rocks. The core is the

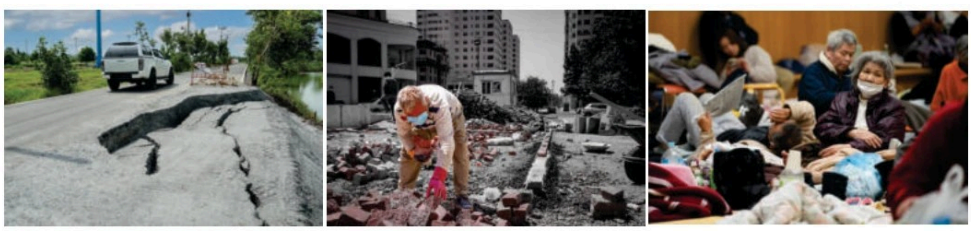
deepest or the innermost layer of the Earth. It is made up of hot, molten material. The core is not in a stable state. A lot of movement goes on inside it. This movement is termed as earthquake.

The point deep inside the Earth where an earthquake originates is called the focus and the point on the Earth's surface directly above the focus is called the epicentre. The maximum effect of the earthquake is felt at the epicentre and its effect reduces as we move away from the epicentre.



Effects of Earthquake

- Destruction of buildings, trees, roads and vehicles
- Huge loss of life (to humans and animals)
- Spread of diseases



Measuring Earthquakes and their Effects

The effects of earthquakes depend on the force with which they occur. Some earthquakes are so mild that they merely shake things kept on a shelf, while some are so strong that they cause huge loss of life and property.

Modern siesmograph

Intensity of an earthquake can be measured using an instrument called the seismograph. Richter scale is a standard scale that is used to find out the intensity of earthquakes on a scale of 1 to 10. Earthquakes measuring 1 to 2 on the scale will hardly be felt. Earthquakes measuring 3, 4 and 5 on the scale can cause damage to property. Earthquakes measuring 6 and above on the scale can result in huge damage to life and property.

Just a Heads up

Zhang Heng of China's Heng dynasty was the first person to make an instrument to detect earthquakes.

Safety Measures during an Earthquake

Here are some steps which can be taken to remain safe during an earthquake.

- Do not run around in panic.
- If you are inside the house, take shelter under a table or a bed.

- Do not stand under or near objects that can fall and hit you.
- Do not use an elevator if you are on the upper floor of a building. Always use the staircase.
- If you are outdoors, stand in an open area away from any building or trees.

Volcanoes

A volcano

A volcano is an opening or a rupture in a heavenly body's surface or crust, which allows hot-molton rocks, ash and gases to escape from below the surface. Deep inside the Earth there are hot-molten rocks called magma. This magma comes up and collects in a chamber. When the pressure inside the chamber builds up, the magma bursts out from a vent or opening or a crack in the crust of the Earth. A volcano is formed when gas, pieces of rocks, ash and lava erupt from vent. When magma erupts through the earth's surface, it is called lava. On cooling, it forms a volcanic rock. The hollow top of the volcano that is shaped like a bowl is called a crater.

There are three types of volcanoes—active, dormant and extinct.

Active volcanoes: These are considered to be highly dangerous. They may erupt at any time or may have erupted in recent years. Some of the most active volcanoes are Mount St. Helens in the United States, Mount Mayon in the Philippines and Mount Lassen in the United States.

Dormant volcanoes: Volcanoes that have not erupted for a long period of time but may erupt in the future are called dormant volcanoes. Some of the dormant volcanoes are Mauna Kea in Hawaii, Mount Edziza in Canada and Narcondam in Andaman Islands of India.

Extinct volcanoes: Volcanoes that are not expected to erupt in the future are called extinct volcanoes. Some of the extinct volcanoes are Mount Ashitaka in Japan, Shiprock in the United States and Zuidwal in the Netherlands.

Just a Heads up

A region in the Pacific Ocean which consists of 452 volcanoes is called the Ring of fire.

Effects of Volcanic Eruption:

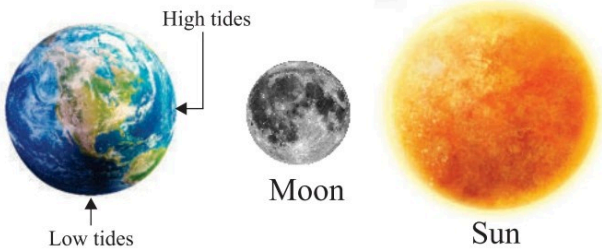
- Destruction of farmlands and roads, damage to property and loss of life.
- Volcanic dust blocks sunlight for a long period, which leads to change in the weather pattern.
- Spread of diseases.

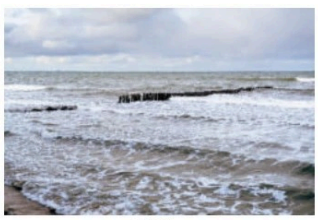
Advantages of Volcanic Eruptions

- After volcanic eruption, the soil becomes very fertile and good for farming.
- Some of the volcanic locations can be used to generate geothermal energy.

Tidal Waves and Tsunami

Tides are the occurrences of rise and fall in the sea level. Sometimes, the waves on the sea are high and sometimes they are low. This rise and fall of the waves is due to the gravitational force of the Sun and the Moon acting on the oceans and seas of the Earth. Tides reach the maximum height when the Sun, Moon and the Earth are in a straight line.




Tides

Tsunami is an enormous wall of water that comes as a series of waves onto the sea shore. These waves have a lot of force and can cover long distances without losing much power and energy. The waves can be as high as 15 metres and travel at a speed of 800 km/hr.

Tsunami Waves

A tsunami can be caused either by an earthquake, volcanic erruption or a land slide. Tsunami waves are so powerful and destructive that they can affect the entire ocean basin.

On December 26, 2004, a Tsunami occurred in India which caused huge loss of life and property in the coastal regions of Tamil Nadu and Kerala. The Andaman and Nicobar islands were the worst hit areas.

Floods

An overflow of a huge amount of water beyond its normal limits over land is called flood. We know that more than 70% of the Earth's surface is covered by water. When the water bodies like oceans, rivers, lakes and ponds have excess water, they overflow, leading to floods.

Flood

Floods occur due to natural factors like:

- Heavy rainfall
- Low-pressure region formed in the ocean and seas

- Excessive melting of snow in the mountains and glaciers

Floods can be prevented if there are:

- Proper streams or channels for the excess water to flow through.
- Enough trees to slow down the force with which water flows.
- Enough mud to absorb excess water.
- Proper embankments that control the level of water flowing into the rivers.

Effects of Floods:

- Loss of life
- Damage to property
- Spread of water-borne diseases

In June 2013, the Indian state of Uttarakhand and nearby places received heavy rainfall, leading to devastating floods and landslides. More than 5700 people lost their lives during this disaster.

Safety Measures during Floods

Here are some of the ways in which we can remain safe during a flood:

- Do not panic. Always listen to flood warnings broadcasted on radio and news channels.
- Abandon the area where the flood water is bound to enter.
- Relocate to higher locations where the flood water cannot reach.
- Keep medicines, food supplies and clean drinking water within your reach.
- Follow the instructions given by rescue team workers.

Droughts

A drought is a condition or situation of severe shortage of water supply (both surface and underground). Some droughts are of shorter duration, whereas some last or continue for years.

Drought

Droughts occur due to lack of or insufficient rainfall along with excessive usage of existing water resources.

Droughts can lead to:

- Destruction of crops.
- Malnutrition and death of humans and animals.
- Mass movement of people from one place to another.

The effects of drought can be controlled by:

- Making judicious usage of available water resources.

Just a Heads up

Drip irrigation is a modern irrigation method that saves water. It is done using narrow tubes which allow very small quantity of water to drip into the roots of plants.

- Adapting to rainwater harvesting.
- Recycling of water.
- Cultivating crops which require less water.

Major drought-prone regions in India are Southern and Eastern Maharashtra, Northern Karnataka, Andhra Pradesh, Orissa, Gujarat and Rajasthan.

In 2012, the Indian state Karnataka faced severe drought.

Make Teaching Fun

Charles Francis Richter was an American scientist who developed the scale, which measures and records the intensity of earthquake. The scale was named after him and is known as 'Richter Scale'.

Words Aid

- **Earthquake:** the sudden movement of a part of the Earth's surface
- **Tide:** the rise and fall of waves in the sea
- **Drought:** a state of severe shortage of water supply (both surface and underground)
- **Flood:** a natural disaster caused by too much rain or inflow of water in an area

Quick Review

- The actual point deep inside the Earth where an earthquake originates is called the focus and the point on the Earth's surface directly above the focus is called the epicentre.
- Intensity of an earthquake can be determined using an instrument called seismograph.
- Volcanoes are of three types, namely active, dormant and extinct

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. The actual point where an earthquake originates is called _____.

   a. mantle b. core

   c. focus d. tide

2. Which of the following natural disasters is caused by shortage of water?

   a. Drought b. Tsunami

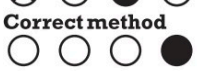
   c. Mantle c. Earthquake

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method



1. (a) (b) (c) (d)
2. (a) (b) (c) (d)

3. Heavy rainfall can cause _____.

a. earthquake b. drought

c. flood d. volcano

3. (a) (b) (c) (d)

B. Fill in the blanks with the correct words from the box :

seismograph, overflow, core, active

1. The innermost layer of the Earth is called _____.
2. Intensity of earthquake is measured using _____.
3. Volcanoes that erupt frequently are called _____ volcanoes.
4. Floods occur due to _____ of water.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Drought occurs when water bodies have excess water to handle.
2. Effect of earthquake is felt more at the epicentre.
3. Tides and Tsunami are one and the same thing.
4. When magma comes out the Earth's surface, it is called lava.

D. Answer these questions :

1. What is an earthquake?
2. What is epicentre?
3. Explain the term dormant volcano.
4. How do tides occur?
5. How do earthquakes occur? What are the aftereffects of an earthquake?
6. What is a volcano? Explain different types of volcanoes with one example of each.
7. Write two aftereffects and two advantages of volcanic eruption.
8. When do floods occur? What are the aftereffects of flood?

Sci-Fun

Name the following :

1. A natural disaster in which the earth suddenly shakes. _____
2. The study of earthquakes. _____
3. The scale on which earthquakes are measured. _____
4. An opening in the crust of the earth. _____
5. A volcano that is not expected to erupt again. _____

Brain Teaser (HOTS)

- **Joy is on the first floor of a building. If an earthquake occurs suddenly, which method is better for him to come out of the building—using the stairs or the elevator? Why?**
- **Can an earthquake cause a Tsunami? Explain.**

Subject Enrichment

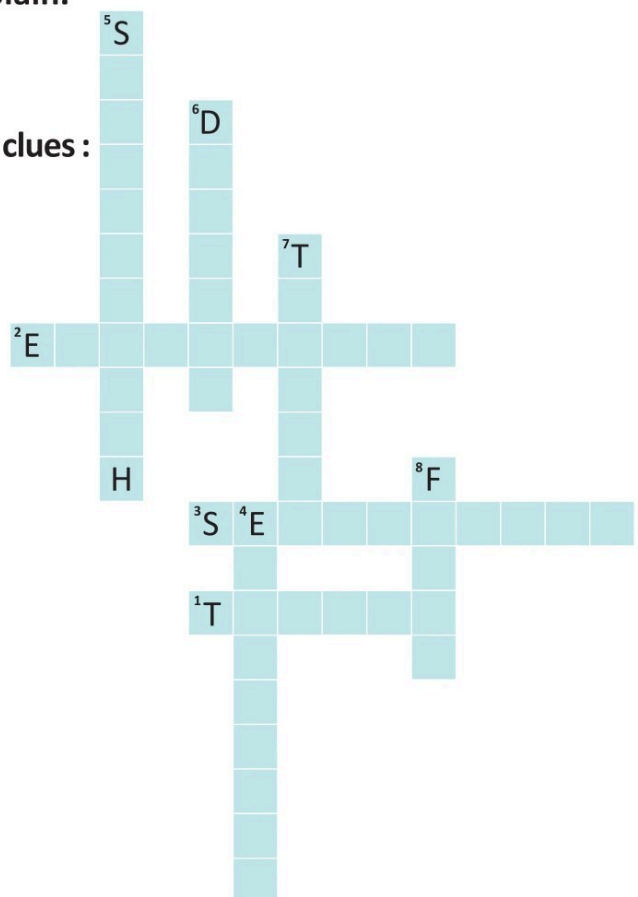
Solve the puzzle with the help of the given clues :

Across

1. Waves caused by the forces of moon
2. Sudden shaking of the earth's surface
3. Study of earthquakes

Down

4. Point on the surface of earth above the focus
5. Instrument to measure earthquake
6. Extreme dry condition
7. Underwater earthquake
8. Point where an earthquake begins



Enhance your Life Skills

Each one of us can help the victims of a natural calamity. We can do this by donating cash or other necessary items to the government, non-government, or charitable organizations.

Subject Connect

Social Studies

On the map of India, mark the regions where volcanoes existed in the recent past. Is there any place on Earth where there is a higher prevalence of volcanoes?

English

Collect information on the safety measures that can be taken during an earthquake. You can take help from the library, newspapers, magazines or the internet and write in your notebook.

13 Our Environment

Intended Learning Outcomes

- What is Environment?
- Pollution
- Greenhouse Effect
- How to Improve the Surroundings
- Destruction of Environment
- Types of Pollution
- Cleanliness of Environment
- Swachh Bharat Abhiyan

Kick Off

We often hear the word 'environment' being used on television, in newspapers and by people around us. Our elders tell us that the 'environment' is not what it used to be earlier; while others say that there are issues related to developed and developing countries. Meeting are regularly held between countries to discuss 'environmental issues. Let's read about the environment.

Funtopia

Observe the pictures given below, and answer the question that follows.

In which of the given places would you like to live and why?

What is Environment?

Various things present around us form our environment. It includes air, water, sunlight, temperature, soil and other living organisms, appropriate for its way of life.

Our environment contains both living things and non-living things. All living things, including human beings, depend on the non-living things. For example, we need air to breathe and water to drink. We get these from the environment around us.

The food that we eat also comes from plants and animals. We obtain raw materials like soils, sand, stones, etc. from the Earth to build our homes and other buildings.

Destruction of Environment

Increase in population and advancement in technology has resulted the change in the environment. To meet the demand of people, green lands have to be changed into high-rise buildings and big industries.

Number of vehicles has significantly increased on the road, which release smoke. Even the factories release harmful smoke. Thus, human activities have also increased considerably. The effects of the human activities are deforestation, degradation of soil, acid rain, greenhouse effect and pollution. Let us know more about pollution and greenhouse effect.

Pollution

The presence of toxic substances which have harmful and poisonous effects on the environment are called pollution.

The substances that cause pollution are called pollutants. Pollution is harmful for plants, animals and human beings.

Cause of Pollution

Human beings have altered the environment over the last hundreds of years. Due to increase in human population more land is being used to provide shelter. Forests are being cut down for various purposes. Large amount of fertilisers are put into the agricultural fields to increase food production. Construction of dams, power plants and other industries has changed the environment adversely. Thus many human activities are not only polluting the environment, but also harming other living organisms.

Types of Pollution

There are various types of pollution.

Air Pollution

Air pollution occurs due to release of pollutants such as dust, smoke, harmful gases from factories, vehicles and from the burning of fossil fuels into air. They lower the quality of air and make it harmful for our health. Breathing polluted or dirty air causes asthma and lung cancer.

Air Pollution

Gases such as carbon dioxide and methane added to the air increase the temperature of the Earth. This results in global warming. Erratic rains and droughts around the world are some of its consequences.

Just a Heads up

Delhi and in some major cities, government has made it mandatory to use CNG (Compressed Natural Gas) in the public vehicles as it causes less pollution compared to petrol and diesel.

Water Pollution

Water Pollution

The introduction of harmful substances in water is called water pollution. This deteriorates the quality of water. Water pollution is caused by the release of effluents from industries containing poisonous substances, disease-causing germs, etc. People often bathe their animals, wash clothes and utensils and even throw household wastes causing the water pollution.

Activity

Collect some water from a nearby water body in a transparent glass. Take some water from a water purifier in another transparent glass. Mark both the glasses as 'A' and 'B' and keep them undisturbed for hours. Now check, what do you notice? Is the water in both the glasses same? In which glass some sediments are seen at the bottom and why?

Land Pollution

Land Pollution

Land pollution is the addition of waste to the land as a result of human activities that degrade the soil and groundwater. Pesticides, fertilisers, acid rain and solid wastes such as polythene bags, bottles and cans cause land pollution. Land pollution makes land infertile.

Noise Pollution

When excessive noise is produced, it causes noise pollution. Some main sources of noise pollution are machines in factories, vehicles, loudspeakers and entertainment gadgets such as television and radio.

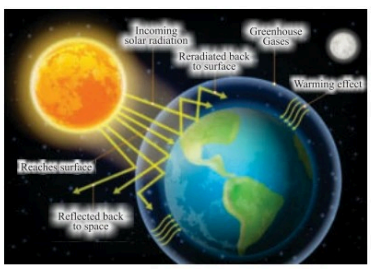
We know that the atmosphere envelops the Earth. When the rays of the Sun fall on the Earth's surface, the surface traps some of the sunrays. The rest of the rays are reflected by the Earth's surface and the atmosphere. The absorbed Sun's energy makes the surface of the Earth warm. This is similar to what happens in a greenhouse.

Just a Heads up

Gases present in the atmosphere that can trap the energy from the Earth are termed as 'greenhouse gases'. The most common greenhouse gases are water vapour, carbon dioxide and methane.

Greenhouse Effect

The Earth is like a glasshouse, also known as greenhouse, where plants are grown in an enclosed space with walls made of glass. These walls allow the Sun rays to enter the glasshouse but do not allow the heat to escape. Therefore, the heat of the Sun warms the inside of the glasshouse creating warm and moist air suitable for plant growth. In a greenhouse, climate is controlled.



Greenhouse effect

The harmful gases emitted by factories and vehicles enter the Earth's atmosphere. These gases include carbon dioxide, methane, nitrous oxide, water vapour and chlorofluorocarbons (CFCs). These gases are able to trap the heat from the Sun and prevent it from escaping the Earth's atmosphere. This results in further heating of the Earth's surface. These gases are called greenhouse gases. The increase in the concentration of greenhouse gases in the atmosphere leads to an increase in the amount of heat close to the Earth's surface. This is called the greenhouse effect.

Greenhouse Gases

The important greenhouse gases are carbon dioxide, methane, nitrous oxide, water vapour, ozone and chlorofluorocarbons (CFCs).

Carbon dioxide

Increased burning of fossil fuels în factories, power plants and vehicles and destruction of rainforests have caused an increase in the concentration of carbon dioxide in the atmosphere over the last hundred years. If the concentration of this gas continues to increase, it will lead to trapping of more heat from the Sun which in turn will result in further increase in the Earth's temperature.

Methane

Methane is added to the atmosphere as a result of the burning of fossil fuels such as coal and petroleum; emissions from natural gas industries and animal farming; and decay and decomposition of waste and wetlands. When the grass-eating animals digest their food, their bodies

release large amount of methane in the atmosphere. The waste of animals, called manure, and fertilisers also release methane into the atmosphere.

Nitrous oxide

The main sources of nitrous oxide are the burning of fossil fuels, and plant and animal matter. In addition, the use of fertilisers in agricultural practices also leads to release of nitrous oxide into the atmosphere.

Water Vapour

Water vapour is the Earth's most abundant greenhouse gas. It is responsible for 66-85% of the greenhouse effect.

Ozone

Ozone at surface level is another significant greenhouse gas. Urbanisation and industrialisation are the major causes of increase in the amount of ozone in the air.

Chlorofluorocarbons (CFCs)

CFCs are chemicals commonly used in refrigerators, air conditioners and industrial processes. It also traps the energy of the Sun and warms up the atmosphere.

Cleanliness of Environment

The environment is our immediate surroundings. It is our moral and social duty to keep our surroundings clean. Our surroundings include our neighbourhood and even the school where we spend lot of time everyday.

Living in clean places has always been good for people. The air we breathe, the water we drink and the roads we walk on must be free from pollution. Hence, it is important to always keep the parks, lawns, roads, and other areas near our homes clean.

What makes your surroundings dirty?

Many reasons that make the surroundings unclean or dirty are, when:

- dirty water is collected in the surroundings.
- mosquitoes and houseflies are seen everywhere.
- people spit on the road.
- plastic items are thrown in the drains.
- drains get clogged because of plastic.

Steps to keep the surroundings clean

- Try to keep your house and the neighbourhood clean.

- Keep the classroom and school premises clean.
- Throw the garbage in dustbins.
- Throw the used polythene bags and plastic bottles in separate bins.
- Clean the dustbin regularly.
- Maintain the toilets in a clean and hygienic condition.
- Never spit on the roads.

How to Improve the Surroundings

There are several ways and means to improve the cleanliness of the surroundings. All of us need to put in a little effort, as cleanliness is possible by the joint efforts of everyone.

- Keep separate bins for plastic and non-plastic wastes.
- Separate the plastic and non-plastic garbage.
- Throw the wastes in the appropriate bin to dispose of the waste properly.
- Recycle and reuse things that you do not use any more.
- Plant more and more trees in your surroundings, they will keep the air fresh.
- Say NO to plastic.
- Use paper or cloth bags for shopping.

To keep the cities and towns clean all over the country, a cleanliness drive has been started by the government of India, which aims to keep the roads, streets and other setups clean. Let us know about it.

Swachh Bharat Abhiyan

Swachh Bharat Abhiyan, also called, the Clean India Mission is a national level campaign initiated by the Government of India to make all the towns clean. This campaign was officially launched on 145th birth anniversary of Mahatma Gandhi on 2nd of October in 2014 at Rajghat, New Delhi.

This movement involves the building of toilets, promoting cleanliness in the rural areas, cleaning streets and roads of the country.

Make Teaching Fun

Chandi Prasad Bhatt is an Indian environmentalist. In 1964 he founded Dasholi Gram Swarajya Sangh in Gopeshwar, which later became a mother-organisation to the Chipko Movement.

He was one of the pioneers in this movement. Today, he is known for his work on environment and considered one of India's first modern environmentalists.

Words Aid

- **environment:** the surroundings in which living and non-living things exist and interact
- **pollution:** the presence of harmful and poisonous substances in the environment
- **pollutants:** the substances that cause pollution

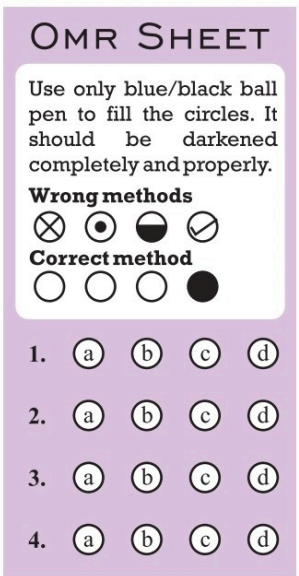
Quick Review

- All that surrounds us is our environment.
- All the organisms depend on the environment for food, shelter, space and water.
- Increase in population and advancement in technology has changed the environment.
- Pollution is the introduction of harmful substances into the environment.
- The main cause of pollution is human activities.
- Addition of harmful substances in air is called air pollution.
- The introduction of harmful substances in water is called water pollution.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. Which of the following is greenhouse gas?
   a. Nitrogen b. Hydrogen
   c. Methane d. Oxygen
2. Which of the following may cause land pollution?
   a. Sewage sludge b. Manure
   c. Water c. Plants
3. Which of the following is harmful for the environment?
   a. Rain b. Wind
   c. Smoke from vehicles d. Growing plants
4. Which of the following is the main cause of pollution?
   a. Human activities b. Growing plants
   c. Discarding plastic d. Using manure



B. Fill in the blanks with the correct words from the box :

environment, clean, human activities, water pollution, land

1. The effects of the _____ _____ are deforestation, degradation of soil, etc.
2. The introduction of harmful substances in water is called _____.
3. Emission of gases from industries into atmosphere pollutes the _____.
4. Deposition of solid waste materials and poisonous chemicals on land is called _____ pollution.
5. *Swachh Bharat Abhiyan* is the _____ India Mission.

C. Think and answer :

1. All that surrounds you _____
2. Cutting down of trees in mass scale _____
3. Introduction of harmful substances into the environment _____
4. Pollution caused due to harmful matter in water _____
5. A greenhouse gas _____

D. Answer the following questions :

1. Define environment.
2. What is global warming?
3. What leads to changes in the environment?
4. What is pollution? Name the different types of pollution.
5. What are greenhouse gases?
6. What do you understand by land pollution?
7. What is greenhouse effect?
8. What steps can you take to improve the cleanliness of the environment?

Sci-Fun

As a student, what can you do to protect the environment, starting from your home?

Brain Teaser (HOTS)

Do you see any changes in weather? If yes, why?

Subject Enrichment

Imagine your father has a small factory in the city. What suggestions can you give him for disposing of the wastes?

Enhance your Life Skills

We can protect our environment by planting trees in the garden, separating plastic and organic waste at home, saving water, switching off unused lights, and helping clean parks, streets, and local surroundings every day.

Subject Connect

Environmental Science

Prepare a list of things of daily use that cause pollution and put it up on your school's bulletin board.

14

Safety Rules and First Aid

Intended Learning Outcomes

- Safety from Fire
- Safety on the Road
- First aid

Kick Off

Accidents happen suddenly and cause physical injury. They usually happen when we are not careful. Doctors say that, if an injured person receives help immediately, it increases his chance of recovery. Let us read about safety and first aid in different situation.

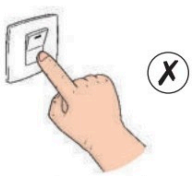
Funtopia

Tick (✓) the picture where safety rules are being followed and cross (✗) the one where safety rules are not being followed:

Safety from Fire

A fire can be caused due to gas leakage, electrical faults or careless handling of candles and matchsticks. We should follow some safety rules to prevent ourselves from fire accidents.

- Switch off the gas leakage, neither strike a match nor operate any electrical switch. Open all doors and windows and call an adult for help.
- Replace faulty gas pipes immediately.
- Check wiring and electrical appliances regularly for any fault or breaks in the wiring.
- Be careful while handling matchbox, candles, firecrackers, etc.



Do not operate electrical switch in case of gas leak.

Do not strike a matchstick in case of gas leak.

- Do not wear synthetic clothes like nylon clothes when handling fire because these clothes catch fire easily.
- If a person's clothes catch fire, they should be rolled onto the floor until the fire is put out. They can also be covered by a blanket to put out fire.

A fire can be put out by the following methods.

Fire extinguisher

- In case of fire accident, immediately call adults for help and inform fire station. The emergency number of a fire station is 112 or 101.
- In case of a small fire, use fire extinguisher to put it out.
- In case of electrical fire, switch off the main switch immediately. Do not use water to put out electrical fire. Use sand, mud or fire extinguisher to put out electrical fire.
- In case of fire caused by petrol, throw mud and sand instead of the water. This is because petrol is lighter than water; it will float up on water and continue burning.

Safety on the Road

We should always be careful on the road. Safety on the road is more important than ever before because of the sudden increase in number of vehicles in the past few years. Being careless on the road can lead to serious accidents. We all should follow the safety rules on the road. On the roadsides, we see many traffic signals. We all should learn about them and follow these signs.

Speed Breaker Ahead

No Entry

School Ahead

No U Turn

Zebra Crossing

Just a Heads up

The very first commercial First Aid kit was made in 1888 by Johnson and Johnson. It was especially made for the railroad workers who would get injured during repair work.

First Aid

It is important that all of us should know what kind of help should be provided immediately if a person around us is hurt. The immediate help given to a sick or injured person before taking him or her to the doctor is called first aid. It can help their condition and provide some relief from pain. At times, it can even save the life of a victim. Let us read about some first aid tips.

First Aid for Burns

First aid on burn

In case of a minor burn, hold the burnt area under cold running water till the burning sensation becomes less. Apply an antiseptic cream or a paste of baking soda and water on the burnt area.

If the burn is serious, immediately call a doctor. Blisters, if formed should not be pricked.

In case of chemical burns, wash the affected area with plenty of water and cover the burn with sterile gauze. Take the victim to a doctor.

First Aid for Animal Bites

First aid on animal bite

Children can be bitten by stray animals such as dogs. An animal scratch can sometimes be as harmful as an animal bite. When these animals bite someone, they release the saliva into the skin of the victim. This saliva can cause a deadly disease called rabies. Hence, it is very important to treat animal bites immediately.

- Wash the wound with soap and water to remove the saliva.
- Apply an antiseptic lotion on the wound to prevent infection.
- Take the victim to a doctor. The doctor may give anti-rabies injection.

In case of snake bite, it injects poison into the victim's body. The poison enters the blood and spreads to all the parts of the body. Do the following in the case of a snake bite.

First aid for snake bite

- Do not move the victim because movement will cause the poison to spread faster.
- Tie a piece of cloth or a bandage near the bite to slow the spread of venom, and take the victim to a doctor. Anti-venom injections may be necessary.

In case of bee stings, its stinger remains inside the skin that causes a lot of pain. Do the following in the case of a bee sting.

- Remove the stinger using tweezers.
- Apply ice on the affected area.
- Apply a paste of baking soda or toothpaste.

First Aid for Fractures

A crack or breakage in a bone is called fracture. The fractured bone causes a lot of pain. The affected area may swell up. You will not be able to move. Following are the steps involved in first-aid provided in case of fracture:

- Do not allow the injured person to move.
- Make the patient calm and comfortable.
- Tie a splint to give support to the broken bone.
- An X-ray of the fractured part shows the extent of injury. The doctor then decides the next course of treatment.

First aid for fracture

Just a Heads up

Sling is a piece of cloth tied around the neck to support the broken arm.

First Aid for Sprain

A sprain is caused when the tissues around a joint get torn. The affected area swells up. On the swelling, rub ice till the swelling subsides.

First Aid for Nose Bleed

At times, when it is very hot, nose may bleed. In such a situation, keep the patient upright with his head held back. Pour cold water on the patient's head. Press the bleeding side of the nose firmly. The patient must not put his finger in the nose.

First aid for nose bleed

Make Teaching Fun

Discuss with the students what might happen to people when safety rules are not followed while doing certain activities. You may ask students to share if they have seen people being careless and not following safety rules. Also guide them how to deal with strangers.

Words Aid

- **Fire extinguisher:** a device containing water or certain other substances that can be used for putting out fire
- **First aid:** the immediate help given to a sick or an injured person
- **Fractured:** a cracked or a broken bone
- **Rabies:** a disease caused in humans due to the bites of infected animals such as dogs and monkeys

Quick Review

- A fire can cause due to gas leakage, electrical faults or careless handling of candles and matchsticks.
- We should follow some safety rules to prevent ourselves from fire accidents.
- We should always be careful on the road.
- The immediate help given to a sick or injured person before taking him or her to the doctor is called first aid.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET:

1. Why do most of the accidents occur?
   a. Due to our carelessness b. Due to our farsightedness
   c. Due to our care d. Due to our fear
2. Which is the common problem among children during hot summer?
   a. Sprain b. Wound
   c. Nose bleeding d. Fracture
3. How can you put out fire caused by the electrical fault?
   a. By petrol b. By fire extinguisher
   c. By water d. By sand or mud
4. A crack or breakage in a bone is called _____.
   a. fracture b. sprain
   c. nose bleeding d. none of these
5. Fire extinguishers are used to put out _____.
   a. mud b. sand
   c. fire d. wood

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1.	(a)	(b)	(c)	(d)
2.	(a)	(b)	(c)	(d)
3.	(a)	(b)	(c)	(d)
4.	(a)	(b)	(c)	(d)
5.	(a)	(b)	(c)	(d)

B. Fill in the blanks with the correct words from the box :

antiseptic, fire, anti-rabies, injured, physical

1. Accidents happen suddenly and cause _____injury.
2. In case of a small_____, use fire extinguisher to put it out.
3. The immediate help given to an _____ person is called first aid.
4. In case of small burns, apply _____ on the burnt area.
5. If an animal bites, the doctor may give _____ injection.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. A sprain is a minor injury and never requires treatment by a doctor.
2. A person suffering from a nose bleed should be asked to breathe fast.
3. The victim of a snake bite should be moved vigorously.
4. The fractured bone causes a lot of pain.
5. In case of chemical burns, wash the affected area with plenty of petrol.

D. Match the following :

1.	Sprains	a.	cover the victim with a blanket and ask him or her to roll on the ground
2.	Fractures	b.	wash the area under running water and apply ice
3.	Burns	c.	apply an ice pack and do not move the joint
4.	Animal bites	d.	use a splint and do not move the bone
5.	Fire	e.	wash the affected area with soap and water, and apply an antiseptic cream

E. Answer the following questions :

1. What is first aid?
2. Differentiate between a sprain and a fracture.
3. What is the use of fire extinguisher?
4. How is rabies caused?
5. How would you help if a person gets snake bite?

Sci-Fun

Look at the image and put a tick mark for the correct activities.

Brain Teaser (HOTS)

Do you think the domestic animals should also be immunised regularly?

Subject Enrichment

Take an old X-ray and try to interpret its findings with the help of the concerned person.

Enhance your Life Skills

Your mother always keeps the medicines at the top shelf, why?

Subject Connect

English

Write a letter to your friend asking whether they were ever stung by a bee. If yes, ask about the preventive measures they took at that time.

Model Test Paper-1

Chapters 1 to 7

A. Darken the correct option in OMR SHEET :

1. Seeds are present inside :
   a. leaves b. fruits
   c. flowers d. roots
2. Which vitamin is essential to prevent night blindness?
   a. Vitamin A b. Vitamin B
   c. Vitamin C d. Vitamin D
3. _____ are lung breathers.
   a. Insects b. Fish
   c. Mammals d. Birds
4. Long bones are filled with _____.
   a. sponge b. simple tissue
   c. bone marrow d. tendon
5. Due to _____ force, a mug floats on water.
   a. elastic b. frictional
   c. buoyant d. axle

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)
3. (a) (b) (c) (d)
4. (a) (b) (c) (d)
5. (a) (b) (c) (d)

B. Fill in the blanks :

1. The person who works on a _____ is known as a farmer.
2. Excessive intake of fatty foods can cause _____.
3. The blood comes to the lungs through tiny blood vessels called _____.
4. The molecules in _____ are loosely packed.
5. Push or pull is called _____.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Some flowers are damaged by heavy rain and strong wind
2. Children need 8-10 hours of sleep in a day.
3. Bones are soft and spongy inside.
4. The sense of smell and taste are interlinked.
5. Water is known as the universal solvent.

D. Answer the following questions :

1. What is reproduction?
2. What is the use of vitamins and minerals in our body?
3. How is the body of the fish adapted to live in water?
4. How is a ball and socket joint different from a gliding joint?
5. What do you understand by reflex action?
6. What are miscible and immiscible liquids?

Model Test Paper-2

Chapters 8 to 14

A. Darken the correct option in OMR SHEET :

1. It improves the rate of sedimentation _____.
   a. distillation b. heating
   c. alum d. filtering
2. Which is the uppermost layer of the earth's crust?
   a. Minerals b. Soil
   c. Wind d. Water
3. Only one of these rocks is a sedimentary rock. Which one?
   a. Shale b. Slate
   c. Granite d. Marble
4. The inner part of the core of the Earth is _____.
   a. woody b. liquid
   c. gas d. solid
5. Heavy rainfall can cause _____.
   a. earthquake b. drought
   c. flood d. volcano

OMR SHEET

Use only blue/black ball pen to fill the circles. It should be darkened completely and properly.

Wrong methods

Correct method

1. (a) (b) (c) (d)
2. (a) (b) (c) (d)
3. (a) (b) (c) (d)
4. (a) (b) (c) (d)
5. (a) (b) (c) (d)

B. Fill in the blanks :

1. We need _____ to breathe.
2. Large deposits of _____ and _____ are present in the soil.
3. _____ rocks form during volcanic eruptions.
4. People who study about heavenly bodies are called _____.
5. The immediate help given to an _____ person is called first aid.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Soil does not support the growth of plants.
2. Microorganims present in soil make it infertile.
3. Granite is a metamorphic rock.
4. Drought occurs when water bodies have excess water to handle.
5. The victim of a snake bite should be moved vigorously.

D. Answer the following questions :

1. What is atmosphere? Why is it important?
2. Define rocks and minerals.
3. Explain how tides occur.
4. What is an earthquake?
5. Write two aftereffects and two advantages of volcanic eruption.
6. How is rabies caused?